SINGULAR GAUSSIAN CONDITIONING IN HILBERT SPACE: SCHUR RESIDUALS, FISHER SCALING, AND CAMERON–MARTIN ENERGY

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ABSTRACT. We investigate infinite-dimensional Gaussian models under singular hyperplane conditioning. As the regularized conditioning strength $\lambda \to \infty$, the conditional laws exit the equivalent-measure manifold of the reference prior due to the Feldman–Hájek dichotomy. To resolve this singularity without invoking unbounded operators, we introduce an inverse-free variational Schur projection that reduces the infinite-dimensional problem to its canonical normal form. We prove that this Schur residual guarantees an exact target variance calibration $\Sigma_{\lambda,kk} = \varepsilon_0/(2\lambda^2)$ alongside a projection residual $R_\lambda = O(\lambda^{-6})$. By defining a local blow-up coordinate $b = \lambda(x_K^* - m_{\lambda,K})$, we demonstrate that the rescaled Fisher tensor admits a Schur-precision expansion. The leading term constitutes a flat Mahalanobis Fisher scaling, while the first nontrivial correction—extracted via a Fenchel-dual variational formulation—is precisely the Cameron–Martin Renormalized Fisher Information Metric (RFIM) residue. When applied to an elliptic Gaussian field, this residue specializes to the Sobolev–Dirichlet energy. These results characterize the exact differential geometry of singular boundaries purely through the algebraic invariants of the Hilbert-space Gaussian condition.

1. INTRODUCTION

Singular hyperplane conditioning arises when one approximates a hard constraint $X_k = x_k^*$ by a calibrated family of linear dynamics with regularization parameter λ, followed by the limit $\lambda \to \infty$. In finite-dimensional Gaussian dynamics, the resulting conditional covariance is described by an ordinary Schur complement. In infinite-dimensional Hilbert-space dynamics, this algebraic description breaks down: stationary covariance operators are trace-class and compact, hence their restrictions do not admit bounded global inverses.

However, when the state space is an infinite-dimensional separable Hilbert space—as arises in stochastic evolution equations [6] and functional linear

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processes [5]—the standard algebraic machinery collapses. The passage from a smooth restoring control of strength λ to the discrete topological cut $\lambda \to \infty$ constitutes a singular perturbation limit [15] in which three fundamental obstructions emerge:

(1) **Unbounded Precision Operators.** The stationary covariance operator is trace-class and compact [6], so its restriction to the free coordinates lacks a bounded global inverse. The Schur complement cannot be defined via operator inversion. Spectral properties of block operator Schur complements have been studied extensively [21, 10], and the variational "shorted operator" perspective [2] addresses positive-operator compressions. However, these results address spectral enclosures and operator shorting without establishing a direct connection to statistical projection residuals under singular Gaussian conditioning.
(2) **Hyperplane Disintegration Singularity.** Conditioning a Radon Gaussian measure on a codimension-1 hyperplane $X_k = x_k^*$ produces a measure supported on a set of prior probability zero [4]. The joint Kullback–Leibler divergence between the exact conditioned law and any non-degenerate prior is $+\infty$.
(3) **Feldman–Hájek Critical Boundary.** The Feldman–Hájek dichotomy [8, 11, 4] implies that two Gaussian measures on a separable Hilbert space are either mutually absolutely continuous or mutually singular. In the Bayesian inverse-problem literature [19], only soft observations with positive-definite noise (which are algebraically equivalent to Tikhonov regularization [9]) preserve equivalence; hard conditioning forces singularity. Regularized Kullback–Leibler approximations for infinite-dimensional measures [16] provide a finite surrogate in the equivalent regime, but the hard coordinate conditioning lies beyond this regime.

To resolve the geometry of the singularity, we construct a tangent blow-up microscope by defining the scaled local coordinate $b \coloneqq \lambda(x_k^* - m_{\lambda,k})$. Instead of ad hoc singularity subtraction, we investigate the limit through this tangent blow-up coordinate geometry. Pulling the Fisher metric back to the regularized singular conditioning boundary reveals an exact hierarchical expansion. By combining the coordinate pull-back Jacobian λ^{-2} with the exact Schur variance calibration $S_{k,\lambda} = \varepsilon_0/(2\lambda^2) - R_\lambda$ and the Woodbury matrix identity, the rescaled joint Fisher metric tensor cleanly bifurcates into distinct analytical strata in the blow-up limit:

(1) **0th-order normal geometry (Displacement Flatness)**: The non-degenerate 0th-order term defines a flat Mahalanobis tangent cone $2\varepsilon_0^{-1}$, structurally preserving the displacement-flatness of Gaussian location families without manufacturing spurious boundary curvature.

(2) **1st-order horizontal metric limit (Schur-precision correction)**: Strikingly, the 1st-order $O(\lambda^{-4})$ perturbative geometry isolates the Cameron–Martin energy of the horizontal lift of the unobserved bulk fibers, yielding exactly the renormalized Fisher information metric (RFIM) limit $\|a_{Ik}\|^2_{Q_0^{-1}}$.

The cancellation mechanism has four layers. First, exact variance normalization fixes the target variance at $\Sigma_{\lambda,kk} = \varepsilon_0/(2\lambda^2)$. Second, the block-decoupling constraint gives an exact block-Lyapunov resolvent for the off-diagonal drift coupling covariance. Third, Cameron–Martin admissibility converts this coupling into an inverse-free variational Schur subtraction of order $O(\lambda^{-6})$, producing the Schur pole $S_{k,\lambda}^{-1} = (2/\varepsilon_0)\lambda^2 + O(\lambda^{-2})$. Fourth, in the Gaussian case, the Dirac-conditioning singularity is removed by a finite-part regularization, leaving the bounded action $\mathfrak{J}_\lambda = b_k^2/\varepsilon_0 + O(\lambda^{-4})$ (exactly zero under decoupled centering). Every technical tool—Cameron–Martin energy, Douglas range inclusion [7], Grümm ideal convergence [18], Radon disintegration, and Feldman–Hájek equivalence—serves this single mechanical chain.

We resolve these singular limits by establishing a unified mathematical hierarchy:

(1) **Projection Residual Scaling Base Theorem:** For any linear stationary process with finite second moments whose covariance satisfies the Lyapunov equation, we prove a Hilbert-space projection residual scaling (Theorem 9) using covariance-energy forms. The target L^2-projection residual, equivalently the LMMSE residual, satisfies $S_{k,\lambda} = \varepsilon_0/(2\lambda^2) - R_\lambda$, meaning its inverse scales as $S_{k,\lambda}^{-1} = (2/\varepsilon_0)\lambda^2 + O(\lambda^{-2})$.

(2) **Gaussian Conditional Extension and Schur-Precision Expansion:** Under the additional assumption of Gaussian Radon laws, we enhance this base result to a conditional analysis. We construct the conditional predictor via the Cameron–Martin space of the bulk covariance, avoiding unbounded inverse operators. Gaussian disintegration yields exact target coordinate zeros under the conditioned evidence. We define the Gaussian renormalized finite-part functional $\mathfrak{J}_\lambda$ using a Dirac mollification scheme, and prove it converges to $b_k^2/\varepsilon_0 + O(\lambda^{-4})$. If the system is decoupled and the evidence is centered, the functional vanishes exactly for all $\lambda > 0$. We then demonstrate that the rescaled joint Fisher metric tensor admits a clean asymptotic expansion where the 0th-order normal geometry is flat and the 1st-order perturbation isolates the Cameron–Martin coupling energy of the off-diagonal drift.

Our approach differs fundamentally from the Bayesian inverse-problem literature [19], in which soft observations with positive-definite noise preserve Gaussian equivalence. We study the measure-theoretic structure of the singular conditioning limit itself—specifically, the collapse of the joint information divergence on the underlying operator algebra as the regularization parameter diverges.

1.1. **Related work and positioning.**

Equivalent-measure manifolds and singular boundaries. The classical extension of information geometry to infinite-dimensional spaces [1, 17, 14] constructs statistical manifolds on the set of probability measures equivalent (i.e., mutually absolutely continuous) to a given reference measure. This equivalent-measure requirement is both natural and restrictive: it guarantees that the Fisher metric is well-defined and the manifold structure is smooth, but it fails precisely when the model enters a mutually singular regime. Hard hyperplane conditioning ($\lambda \to \infty$) produces such a regime: the conditioned measures converge to a Dirac distribution on the target coordinate and exit the equivalence class of the reference Gaussian law. The tangent blow-up coordinate extracts the rescaled metric limit from the divergent KL boundary, acting as a renormalization at the edge of the equivalent-measure manifold.

The Feldman–Hájek measure-class wall. The Feldman–Hájek theorem [8, 11, 4] establishes a sharp dichotomy for Gaussian measures on separable Hilbert spaces: two Gaussian laws are either equivalent or mutually singular, with no intermediate case. In Bayesian inverse problems, [19] shows that positive observational noise keeps the posterior in the equivalent regime; zero-noise hard conditioning exits that regime. The standard workaround is to force additive observation noise into the model, preserving absolute continuity at the cost of never reaching the exact conditioning limit. Our construction takes a different route: the tangent blow-up isolates the non-degenerate geometric limit while preserving the Cameron–Martin energy structure of the coupling, without injecting artificial noise.

Transport geometry and Bures–Wasserstein limits. Infinite-dimensional Gaussian measures that are mutually singular can still be compared via optimal transport distances. The Wasserstein geometry of Gaussian measures [20] and the Bures–Wasserstein distance between covariance operators [3] provide global metrics that do not require absolute continuity; for background on optimal transport and gradient flows in metric spaces, see [22]. The geometry developed here is complementary: the tangent blow-up metric governs the boundary normal geometry and horizontal lift of singular evidence values (Section 5), while the Bures–Wasserstein metric governs bulk covariance relaxation. The present paper does not use any Wasserstein gradient-flow identity.

Distinction from singular learning theory. [23] develops a theory of singular statistical models in which the Fisher information matrix degenerates due to parameter redundancy (the parameter-to-law map has an algebraic singularity). The singularity studied here is fundamentally different: the parameterization remains linear and the model Gaussian, but the limiting measures leave the common absolutely continuous class under hard conditioning. Watanabe's tools—resolution of singularities, real log-canonical thresholds, learning coefficients—address algebraic degeneracy of the Fisher matrix on a finite-dimensional parameter space. Our tangent blow-up addresses the measure-theoretic singularity of a KL divergence that diverges because the conditioned measures become mutually singular in infinite dimensions.

1.2. Organization of the Paper.

We organize the paper as follows. Section 2 defines the operator model and the variance normalization. Section 3 establishes the Hilbert-space Schur residual scaling. Section 4 details the Gaussian conditional extension and the renormalized finite-part functional. Section 5 establishes the rescaled Fisher tensor and the Schur-precision tangent blow-up expansion. Section 6 demonstrates the framework on an infinite-dimensional Gaussian field model, interpreting the RFIM as a Dirichlet connection energy. Section 7 discusses non-Gaussian limitations. Section 8 concludes. Proofs are given in Sections A to C; numerical diagnostics and Galerkin verification in Section D.

2. Operator Model and Exact Variance Normalization

2.1. Operator Dynamics and Regularized Conditioning.

We model the continuous-time dynamics as stochastic evolution on a separable Hilbert space $\mathcal{H}$ [6]:

$$dX_t = \mathcal{A}X_t\,dt + \mathcal{D}^{1/2}dW_t \tag{1}$$

where $\mathcal{A}$ is the drift operator and $\mathcal{D}$ is the trace-class diffusion operator. We decompose the Hilbert space as $\mathcal{H} = \mathcal{H}_k \oplus \mathcal{H}_I$, where $\mathcal{H}_k := \mathbb{R}\phi_k$ is the one-dimensional target coordinate subspace, and $\mathcal{H}_I := \mathcal{H}_k^\perp$ is the free bulk subspace. Let $\Pi_k := \phi_k \otimes \phi_k$ and $\Pi_I := I - \Pi_k$ be the corresponding orthogonal projections.

To approximate the exact point conditioning $X_k = x_k^*$, we sever all incoming target-coordinate edges and apply a regularization parameter $\lambda > 0$.

Definition 1 (Calibrated Approximating Dynamics Family)**.** The regularized conditioning operator of strength $\lambda > 0$ at node X_k acts on the operator dynamics (1) by modifying both the drift and the diffusion:

$$\mathcal{A}^{(\lambda)} := \mathcal{A}_{\mathrm{cond}} - \lambda\Pi_k, \qquad \mathcal{D}^{(\lambda)} := \mathcal{D}_{\mathrm{cond}} + d_\lambda\Pi_k \tag{2}$$

where $\mathcal{A}_{\mathrm{cond}}$ is the block drift operator with all incoming target-coordinate edges severed ($\Pi_k\mathcal{A}_{\mathrm{cond}} = 0$), $\mathcal{D}_{\mathrm{cond}}$ has target-coordinate noise severed, and $d_\lambda > 0$ is the local diffusion variance of the target coordinate at regularization strength λ. In the SDE, this corresponds to:

$$dX_{k,t} = -\lambda(X_{k,t} - x_k^*)dt + \sqrt{d_\lambda}\,dW_{k,t}. \tag{3}$$

2.2. **Exact Variance Normalization.** The block-decoupled structure under the regularized dynamics yields:

$$\mathcal{A}^{(\lambda)} = \begin{pmatrix} -\lambda & 0 \\ a_{Ik} & \mathcal{A}_{II} \end{pmatrix}, \qquad \mathcal{D}^{(\lambda)} = \begin{pmatrix} d_\lambda & 0 \\ 0 & D_{II} \end{pmatrix}, \tag{4}$$

where $a_{Ik} := \mathcal{A}_{Ik}\phi_k \in \mathcal{H}_I$ represents the off-diagonal drift coupling. Let Σ_λ denote the stationary covariance operator under regularization strength λ, solving the Lyapunov equation:

$$\mathcal{A}^{(\lambda)}\Sigma_\lambda + \Sigma_\lambda(\mathcal{A}^{(\lambda)})^* + \mathcal{D}^{(\lambda)} = 0. \tag{5}$$

The target block kk decouples completely from the free coordinates, yielding the exact scalar equation:

$$-2\lambda\Sigma_{\lambda,kk} + d_\lambda = 0 \quad \Longrightarrow \quad \Sigma_{\lambda,kk} = \frac{d_\lambda}{2\lambda}. \tag{6}$$

Proposition 2 (Exact Variance Normalization)**.** *To maintain a physical target noise scale $\varepsilon_0 > 0$ independent of the auxiliary parameter λ under the singular limit, we define the exact variance normalization condition:*

$$\lambda^2\Sigma_{\lambda,kk} \equiv \frac{\varepsilon_0}{2} \quad \textit{for all } \lambda > 0, \tag{7}$$

which uniquely requires:

$$d_\lambda = \frac{\varepsilon_0}{\lambda} \quad \textit{for every } \lambda > 0. \tag{8}$$

Substitution of this target diffusion scaling yields the calibrated approximating dynamics:

$$dX_{k,t} = -\lambda(X_{k,t} - x_k^*)dt + \sqrt{\varepsilon_0/\lambda}\, dW_{k,t}. \tag{9}$$

All subsequent results are stated for this calibrated family.

Definition 3 (Schur LMMSE Invariant)**.** Let $S_{k,\lambda} := \|X_k - P_{\mathcal{M}_I}X_k\|_{L^2}^2$ represent the linear minimum mean squared error (LMMSE) residual of the target coordinate X_k on the closed linear span $\mathcal{M}_I$ of the free coordinates X_I. The *Schur LMMSE invariant* m_2^{Schur} is defined as:

$$m_2^{\text{Schur}} := \lim_{\lambda\to\infty} \lambda^2 S_{k,\lambda}. \tag{10}$$

2.3. **Operator Setup and Assumptions.** We state the operator-theoretic assumptions that make the limit passage independent of the Galerkin dimension.

Definition 4 (Drift Operator with Volterra Perturbation)**.** A drift operator on a separable Hilbert space $\mathcal{H}$ is an operator of the form $\mathcal{A} = \mathcal{A}_0 + \mathcal{K}$, where:

(i) $\mathcal{A}_0 : \mathrm{dom}(\mathcal{A}_0) \subset \mathcal{H} \to \mathcal{H}$ is a closed, densely defined, dissipative operator that generates a strongly continuous contraction semigroup $(e^{t\mathcal{A}_0})_{t\geq 0}$ on $\mathcal{H}$, satisfying $\|e^{t\mathcal{A}_0}\| \leq e^{-\alpha t}$ for some spectral gap $\alpha > 0$.

(ii) $\mathcal{K} : \mathcal{H} \to \mathcal{H}$ is a compact Volterra operator with spectral radius $\rho(\mathcal{K}) = 0$.

(iii) The full operator $\mathcal{A} = \mathcal{A}_0 + \mathcal{K}$ generates a strongly continuous semigroup $(e^{t\mathcal{A}})_{t\geq 0}$ by the bounded perturbation theorem, and this semigroup remains exponentially stable: $\|e^{t\mathcal{A}}\| \leq Me^{-\beta t}$ for some $M \geq 1$, $\beta > 0$.

In finite dimensions ($\mathcal{H} = \mathbb{R}^n$), $\mathcal{A}_0$ reduces to the Hurwitz drift matrix A and $\mathcal{K} = 0$.

Definition 5 (Trace-Class Noise and Covariance)**.** Let $\mathcal{S}_1(\mathcal{H})$ denote the Banach space of trace-class (nuclear) operators on $\mathcal{H}$ equipped with the trace norm $\|T\|_{\mathcal{S}_1} := \mathrm{Tr}(|T|) = \sum_{i=1}^{\infty}\langle \phi_i, (T^*T)^{1/2}\phi_i\rangle < \infty$.

(i) The diffusion operator $\mathcal{D} : \mathcal{H} \to \mathcal{H}$ is positive, self-adjoint, and belongs to $\mathcal{S}_1(\mathcal{H})$.

(ii) Assuming the pair $(\mathcal{A}, \mathcal{D}^{1/2})$ is controllable, the stationary covariance operator $\Sigma : \mathcal{H} \to \mathcal{H}$ is the unique strictly positive, self-adjoint, trace-class solution to the operator Lyapunov equation:

$$\mathcal{A}\Sigma + \Sigma\mathcal{A}^* + \mathcal{D} = 0 \tag{11}$$

which is given by the convergent Bochner integral

$$\Sigma = \int_0^\infty e^{t\mathcal{A}}\mathcal{D}e^{t\mathcal{A}^*}\,dt.$$

Assumption 6 (Core Analytical Assumptions for Singular Hyperplane Conditioning)**.** This work relies on two core analytical assumptions regarding the target-coordinate cut and the outgoing coupling:

(i) **Linear Second-Moment Channel:** A centered linear stationary process on $\mathcal{H} = \mathcal{H}_k \oplus \mathcal{H}_I$ has finite second moments and covariance solving Equation (11). The block-decoupling constraint zeroes all incoming target drift and cross-noise, so that $\mathcal{A}_{kI} = \mathcal{D}_{kI} = \mathcal{D}_{Ik} = 0$ and $\mathcal{A}_{kk} = -\lambda$. The target diffusion $\mathcal{D}_{kk} = \varepsilon_0/\lambda$ is the unique scaling determined by the exact variance normalization (Proposition 2). The free noise is independent of the local target noise. (Gaussianity is not assumed).

(ii) **Cameron–Martin Stability of Off-Diagonal Drift Coupling:** Let $Q_0 := \Sigma_{II}^{(0)}$ denote the free-block covariance produced by the free noise under block decoupling, and let $H_{Q_0} := \mathrm{Range}(Q_0^{1/2})$ with the energy norm $|u|_{Q_0} := \|Q_0^{-1/2}u\|_{\mathcal{H}_I}$. The outgoing vector $a_{Ik} := \mathcal{A}_{Ik}\phi_k$ belongs to H_{Q_0}, and $e^{t\mathcal{A}_{II}}$ restricts to a strongly continuous exponentially stable semigroup on H_{Q_0}:

$$\|e^{t\mathcal{A}_{II}}\|_{\mathcal{L}(H_{Q_0})} \leq Me^{-\omega t}, \qquad t \geq 0, \tag{12}$$

for constants $M \geq 1$ and $\omega > 0$ independent of λ.

Intuitive Explanation: Part (i) states that the regularized dynamics act as a strong localized drift penalization on the target coordinate X_k while isolating it from incoming influences, representing a physical realization of the block-decoupling constraint. Part (ii) requires that the off-diagonal drift coupling a_{Ik} from the target coordinate has finite energy relative to the background fluctuations of the free coordinates. This ensures that target fluctuations propagating downstream do not overload the variance of the infinite-dimensional system.

Remark 7 (Finite-Energy Outgoing Covariance Coupling and Finite-Dimensional Calibration). The membership $a_{Ik} \in H_{Q_0}$ is an admissibility condition, not a consequence of the free-block Lyapunov equation alone: an arbitrary outgoing vector need not lie in $\mathrm{Range}((\Sigma_{II}^{(0)})^{1/2})$. It says that the deterministic channel carrying target fluctuations into the free-block has finite covariance energy relative to the free-noise background.

To understand this condition intuitively, consider a finite-dimensional setting where the free-block state space is $\mathcal{H}_I = \mathbb{R}^{d-1}$ and the free stationary covariance $Q_0 := \Sigma_{II}^{(0)}$ is strictly positive-definite. Since $Q_0 \succ 0$, its square root $Q_0^{1/2}$ is invertible, and its range is the entire space $\mathbb{R}^{d-1}$. In this case, the Cameron–Martin space is the full space $H_{Q_0} = \mathrm{Range}(Q_0^{1/2}) = \mathbb{R}^{d-1}$, and the Cameron–Martin energy norm $|u|_{Q_0} = \|Q_0^{-1/2}u\|_2$ is equivalent to the standard Euclidean norm. Consequently, the admissibility condition $a_{Ik} \in H_{Q_0}$ is trivially satisfied for any outgoing vector $a_{Ik} \in \mathbb{R}^{d-1}$.

However, if some direction in the free coordinates is noise-free under block decoupling (meaning Q_0 is positive semidefinite but singular), H_{Q_0} becomes a proper subspace of $\mathbb{R}^{d-1}$. In this case, the condition $a_{Ik} \in H_{Q_0}$ requires that the leakage vector a_{Ik} has no component in the null space of Q_0. Physically, it prevents the target coordinate from leaking fluctuations into a downstream coordinate that has zero background noise, which would otherwise result in infinite relative covariance energy.

In infinite dimensions, $\Sigma_{II}^{(0)}$ is trace-class and compact, so its eigenvalues decay to zero, making $\mathrm{Range}((\Sigma_{II}^{(0)})^{1/2})$ a dense but strictly proper subspace of $\mathcal{H}_I$. The condition $a_{Ik} \in H_{Q_0}$ requires the off-diagonal drift coupling coefficients to decay sufficiently fast relative to the background noise eigenvalues to keep the relative energy finite.

Under this assumption, invariance and exponential stability on H_{Q_0} imply:

$$
\begin{aligned}
&(\lambda I - \mathcal{A}_{II})^{-1} a_{Ik} \in H_{Q_0}, \\
&|(\lambda I - \mathcal{A}_{II})^{-1} a_{Ik}|_{Q_0} \leq \frac{M}{\lambda + \omega} |a_{Ik}|_{Q_0}.
\end{aligned}
\tag{13}
$$

Remark 8 (Constant Dependency in LMMSE Expansion). Consequently, the exact leakage cross-covariance $\Sigma_{\lambda,Ik}$ has finite Cameron–Martin energy norm and is $O(\lambda^{-3})$ in that norm. This is a condition on the deterministic response and covariance energy; it does not assert that infinite-dimensional Gaussian sample paths lie in H_{Q_0} almost surely.

3. Hilbert-Space Variational Schur Residual

3.1. **Variational Schur Subtraction and Energy Bounds.** In finite dimensions, the Schur complement of the bulk covariance block Σ_{II} is defined as $S_k := \Sigma_{kk} - \Sigma_{kI}\Sigma_{II}^{-1}\Sigma_{Ik}$. Under stable and controllable dynamics, the bulk covariance Σ_{II} is positive-definite and invertible, so the subtraction is well-defined. However, in infinite dimensions, the stationary covariance operator is trace-class and compact, meaning its restriction $\Sigma_{\lambda,II}$ lacks a bounded global inverse. To lift the Schur complement to infinite dimensions, we formulate the projection residual directly using covariance-energy forms.

Let $Q_0 := \Sigma_{II}^{(0)}$ denote the fixed background bulk covariance under full decoupling ($\lambda = \infty$). We define the core Cameron–Martin space $H_{Q_0} := \mathrm{Range}(Q_0^{1/2})$ and the core energy norm:

$$|u|_{Q_0} := \|Q_0^{-1/2}u\| \quad \text{for all } u \in H_{Q_0}. \tag{14}$$

By Assumption 6, the off-diagonal drift coupling vector satisfies $a_{Ik} \in H_{Q_0}$, and the restricted drift generator $\mathcal{A}_{II}$ generates an exponentially stable semigroup on H_{Q_0}.

Under this assumption, the exact cross-covariance resolvent formula (derived in Section A) satisfies:

$$\Sigma_{\lambda,Ik} = \frac{\varepsilon_0}{2\lambda^2}(\lambda I - \mathcal{A}_{II})^{-1}a_{Ik}. \tag{15}$$

This resolvent response yields the Cameron–Martin energy bound on the cross-covariance block:

$$|\Sigma_{\lambda,Ik}|_{Q_0} = O(\lambda^{-3}) \quad \text{as } \lambda \to \infty. \tag{16}$$

To define the infinite-dimensional projection residual, we introduce the variational Schur subtraction R_λ:

$$R_\lambda := \sup_{\langle u,\Sigma_{\lambda,II}u\rangle>0} \frac{|\langle u, \Sigma_{\lambda,Ik}\rangle|^2}{\langle u, \Sigma_{\lambda,II}u\rangle}. \tag{17}$$

Since the target noise channel and bulk noise channel are independent, the bulk state decomposes into independent background and target-driven components, meaning the bulk covariance dominates the background covariance ($\Sigma_{\lambda,II} = Q_0 + \mathrm{Cov}(Y_I) \succeq Q_0$). Douglas range inclusion then guarantees that the variational subtraction is bounded:

$$0 \le R_\lambda \le |\Sigma_{\lambda,Ik}|_{Q_0}^2 = O(\lambda^{-6}). \tag{18}$$

3.2. **Hilbert-Space Schur Residual Scaling.** We now establish the main scaling theorem for the L^2-projection residual.

Theorem 9 (Hilbert Schur Residual Invariant)**.** *Under Assumptions 6 and 6, the L^2-projection residual satisfies:*

$$S_{k,\lambda} := \|X_k - P_{\mathcal{M}_I}X_k\|_{L^2}^2 = \frac{\varepsilon_0}{2\lambda^2} - R_\lambda, \tag{19}$$

where the remainder satisfies $0 \leq R_\lambda = O(\lambda^{-6})$*. In particular, the Schur LMMSE invariant is:*

$$m_2^{\mathrm{Schur}} = \lim_{\lambda\to\infty} \lambda^2 S_{k,\lambda} = \frac{\varepsilon_0}{2}, \tag{20}$$

and the inverse projection residual scaling satisfies:

$$S_{k,\lambda}^{-1} = \frac{2}{\varepsilon_0}\lambda^2 + O(\lambda^{-2}). \tag{21}$$

This result depends only on covariance and orthogonal projection; it holds for any linear stationary model satisfying these assumptions, without Gaussianity.

Proof. See Section A for the full derivation. □

3.3. **Convergence of Finite-Dimensional Truncations.**

Let $\Pi_n : \mathcal{H} \to \mathcal{H}_n := \mathrm{span}\{\phi_1, \ldots, \phi_n\}$ be the n-dimensional Galerkin projection. Write $\mathcal{A}_n^{(\lambda)} := \Pi_n \mathcal{A}^{(\lambda)} \Pi_n$, $\mathcal{D}_n^{(\lambda)} := \Pi_n \mathcal{D}^{(\lambda)} \Pi_n$, and let $\Sigma_{n,\lambda}$ solve the truncated Lyapunov equation $\mathcal{A}_n^{(\lambda)} \Sigma_{n,\lambda} + \Sigma_{n,\lambda} (\mathcal{A}_n^{(\lambda)})^* + \mathcal{D}_n^{(\lambda)} = 0$.

Proposition 10 (Galerkin Consistency of the Schur Invariant)**.** *Under Assumption 6, let* $\Pi_n : \mathcal{H} \to \mathcal{H}_n$ *be finite-rank orthogonal Galerkin projections preserving the target-bulk block decomposition:*

$$\Pi_n \phi_k = \phi_k, \qquad \Pi_n \mathcal{H}_I \subset \mathcal{H}_I, \qquad \Pi_n \to I \ \textit{strongly on}\ \mathcal{H}. \tag{22}$$

Assume the truncated regularized dynamics satisfy the uniform exponential stability and trace-class conditions stated in Proposition 20 of Section A. Then the truncated covariance operators converge in trace norm:

$$\|\Sigma_{n,\lambda} - \Sigma_\lambda\|_{\mathcal{S}_1} \to 0 \quad \textit{as } n \to \infty, \tag{23}$$

and the truncated Schur invariant is exact at every level:

$$m_2^{\mathrm{Schur},(n)} := \lim_{\lambda\to\infty} \lambda^2 S_{k,\lambda}^{(n)} = \frac{\varepsilon_0}{2} \quad \textit{for all } n \geq k. \tag{24}$$

Proof. See Section A for the dynamic and spectral support. □

4. Gaussian Conditional Extension and Finite-Part Functional

This section gives the Gaussian extension of the base theorem, using Radon Gaussian laws and disintegration to formulate conditioning under exact evidence and define the renormalized finite-part functional.

4.1. **Radon Gaussian Setup and Disintegration.**

Assumption 11 (Hilbert Gaussian Conditioning and Affine Mean Channel)**.** In addition to Assumptions 6 and 6, let $\mu_\lambda = \mathcal{N}(m_\lambda, \Sigma_\lambda)$ be a Gaussian Radon law on $\mathcal{H}$ with injective trace-class covariance. Let μ_{obs} be the canonical Gaussian disintegration member on the evidence hyperplane $X_k = x_k^*$.

Suppose that a fixed deterministic forcing $b \in \mathcal{H}$ gives the stationary mean equation:

$$\mathcal{A}_{\mathrm{cond}} m_\lambda + b - \lambda \Pi_k (m_\lambda - x_k^* \phi_k) = 0, \quad \Pi_k \mathcal{A}_{\mathrm{cond}} = 0. \tag{25}$$

We define the target deterministic offset constant:

$$c_\lambda := \lambda(m_{\lambda,k} - x_k^*) = b_k, \tag{26}$$

where $m_{\lambda,k} := \langle m_\lambda, \phi_k \rangle$ and $b_k := \langle b, \phi_k \rangle$.

Under the Gaussian Radon law, the coordinate projection $\pi_k : \mathcal{H} \to \mathbb{R}$ is continuous, so the target variable X_k is a well-defined Borel random variable. The Polish structure of the Hilbert space guarantees the existence of a regular conditional probability distribution. By selecting the weakly continuous version of the conditional law at the single point $t = x_k^*$ (as detailed in Section B), we obtain the unique pointwise disintegration member μ_{obs}. Under this measure, the target coordinate satisfies the exact target zeros:

$$m_{\mathrm{obs},k} - x_k^* = 0, \quad \mathrm{Var}_{\mathrm{obs}}(X_k) = 0 \quad \text{(exact)}. \tag{27}$$

4.2. **Global-Inverse-Free Measurable Predictor.** Let $C_\lambda := \Sigma_{\lambda,II}$ denote the Gaussian bulk covariance operator on $\mathcal{H}_I$. Because C_λ is trace-class and compact, it lacks a bounded global inverse. We define the bulk Cameron–Martin space $H_{C_\lambda} := \mathrm{Range}(C_\lambda^{1/2})$ and the corresponding energy norm.

The leakage cross-covariance vector satisfies $\Sigma_{\lambda,Ik} \in H_{C_\lambda}$ with energy norm $\|\Sigma_{\lambda,Ik}\|_{H_{C_\lambda}} = O(\lambda^{-3})$. This range inclusion allows us to define the conditional predictor $\mu_{k|I}(X_I) := \mathbb{E}[X_k \mid X_I]$ as a measurable Wiener functional:

$$\mu_{k|I}(X_I) = m_{\lambda,k} + W_{\Sigma_{\lambda,Ik}}(X_I - m_{\lambda,I}), \tag{28}$$

which is the unique LMMSE predictor in L^2. The estimates in Section B imply that, for all sufficiently large λ, the bulk marginals $\mu_{\mathrm{obs},I}$ and $\mu_{\lambda,I}$ are equivalent and have finite bulk relative entropy:

$$D_{\mathrm{KL}}(\mu_{\mathrm{obs},I} \parallel \mu_{\lambda,I}) = O(\lambda^{-4}) \to 0 \quad \text{as } \lambda \to \infty. \tag{29}$$

4.3. **Gaussian Renormalized Finite-Part Functional.** Although the joint KL divergence diverges under the exact conditioning, we can define a renormalized finite-part action.

Definition 12 (Gaussian Renormalized Finite-Part Functional)**.** Whenever $S_{k,\lambda} > 0$ and $\mu_{\mathrm{obs},I}$ is equivalent to $\mu_{\lambda,I}$, we define the Gaussian renormalized finite-part functional of the conditioned law:

$$\begin{aligned}\mathfrak{J}_\lambda(\mu_{\mathrm{obs}}; \mu_\lambda) :=& D_{\mathrm{KL}}(\mu_{\mathrm{obs},I} \parallel \mu_{\lambda,I}) \\ &+ \frac{1}{2} S_{k,\lambda}^{-1} \mathbb{E}_{\mu_{\mathrm{obs}}}\left[\left(X_k - \mu_{k|I}(X_I)\right)^2\right].\end{aligned} \tag{30}$$

This represents the finite part of the joint KL divergence under the conditional normalization scheme, where the Dirac and Gaussian normalization singularities are subtracted.

The finite part is not claimed to be a canonical KL divergence between mutually singular measures; it is the finite part associated with the Dirac-mollification and conditional-normalization scheme above.

4.4. **Gaussian Main Theorem.** We now establish the main regularization and cancellation theorems for the finite-part functional.

Theorem 13 (Gaussian Finite-Part Regularization)**.** *Under Assumption 11, there exists $\lambda_0 > 0$ such that, for $\lambda \geq \lambda_0$, the renormalized finite-part functional under the conditioned law μ_{obs} is well-defined and satisfies:*

$$\mathfrak{J}_\lambda(\mu_{\mathrm{obs}}; \mu_\lambda) = \frac{b_k^2}{\varepsilon_0} + O(\lambda^{-4}). \tag{31}$$

In particular, the functional remains bounded at $O(1)$ and converges to:

$$\lim_{\lambda \to \infty} \mathfrak{J}_\lambda(\mu_{\mathrm{obs}}; \mu_\lambda) = \frac{b_k^2}{\varepsilon_0}. \tag{32}$$

Proof. See Section B for the full derivation. □

Corollary 14 (Exact Decoupled-Centered Cancellation)**.** *Under Assumption 11, if the target coordinate is dynamically decoupled from the bulk ($a_{Ik} = 0$) and the evidence is centered at the stationary mean ($x_k^* = m_{\lambda,k}$), then:*

$$\mathfrak{J}_\lambda(\mu_{\mathrm{obs}}; \mu_\lambda) = 0 \quad \textit{for every } \lambda > 0. \tag{33}$$

Proof. The decoupling implies $\Sigma_{\lambda,Ik} = 0$, so $D_{\mathrm{KL}}(\mu_{\mathrm{obs},I} \parallel \mu_{\lambda,I}) = 0$. The centering makes the conditional residual zero under the conditioning, so both terms in (30) vanish. □

5. Rescaled Fisher Tensor and Schur-Precision Expansion

This section constructs the unified blow-up geometry by combining the coordinate pull-back Jacobian with a Woodbury matrix expansion.

5.1. **The blow-up coordinate and pull-back Jacobian.** Section 3 establishes the exact target variance normalization $\Sigma_{\lambda,KK} = E_0/(2\lambda^2)$ for a finite-dimensional target block of dimension m, while the Schur projection residual satisfies $R_\lambda = O(\lambda^{-6})$. Let $S_{K,\lambda} = E_0/(2\lambda^2) - R_\lambda$ be the target Schur complement matrix.

For two conditional Gaussian distributions with the same covariance matrix, their divergence is precisely given by the Mahalanobis quadratic form of their mean difference. In the original macroscopic physical coordinates x_K^*, this joint Fisher metric is simply $S_{K,\lambda}^{-1}$, which diverges as $O(\lambda^2)$.

To resolve this singular collapse, we define the blow-up coordinate system:

$$b = \lambda(x_K^* - m_{\lambda,K}) \in \mathbb{R}^m. \tag{34}$$

Equivalently, $x_K^* = m_{\lambda,K} + \lambda^{-1} b$. Under the calibrated dynamics, the scaled coordinate b is $O(1)$.

Definition 15 (Rescaled joint Fisher metric tensor)**.** Let $b, b' \in \mathbb{R}^m$ be two blow-up conditioning coordinates. Applying the coordinate pull-back Jacobian, the blow-up divergence at strength λ is

$$\mathfrak{D}_\lambda^{\mathrm{FP}}(b, b') := \frac{1}{2}(b - b')^\top \left[\lambda^{-2} S_{K,\lambda}^{-1}\right](b - b'). \tag{35}$$

The rescaled joint Fisher metric tensor on the blow-up coordinate chart is precisely:

$$g_\lambda^{\mathrm{joint},b} := \lambda^{-2} S_{K,\lambda}^{-1}. \tag{36}$$

5.2. **Woodbury–Schur hierarchical expansion.** We utilize the fully precise Schur calibration decomposition:

$$S_{K,\lambda} = A_\lambda - R_\lambda, \qquad A_\lambda = \frac{E_0}{2\lambda^2}. \tag{37}$$

The matrix R_λ is the inverse-free variational Schur subtraction $R_\lambda = \sup_{u \in \mathcal{H}_I} \{2\langle c_\lambda, u\rangle - \langle u, B_\lambda u\rangle\}$, where $c_\lambda = \Sigma_{\lambda,IK}$ and $B_\lambda = \Sigma_{\lambda,II}$. Under Cameron–Martin admissibility, $R_\lambda = O(\lambda^{-6})$.

Applying the Woodbury matrix identity (or equivalently, the exact Taylor geometric series for inverses), we expand the full covariance inverse operator:

$$S_{K,\lambda}^{-1} = A_\lambda^{-1} + A_\lambda^{-1} R_\lambda A_\lambda^{-1} + \mathcal{O}(R_\lambda^2). \tag{38}$$

5.3. **Unified Blow-Up Metric Theorem.**

Theorem 16 (The Asymptotic Unified Blow-Up Metric)**.** *Applying the coordinate pull-back Jacobian λ^{-2}, the full regularized Fisher geometry cleanly bifurcates into distinct analytical strata in the blow-up limit:*

$$g_\lambda^{\mathrm{joint},b} = \underbrace{2E_0^{-1}}_{\textit{0th-order normal geometry}} + \lambda^{-4} \underbrace{\left[(2E_0^{-1})(\lambda^6 R_\lambda)(2E_0^{-1})\right]}_{\textit{1st-order horizontal metric limit}} + \mathcal{O}(\lambda^{-8}). \tag{39}$$

If the Schur residue has the sharp expansion $R_\lambda = C_R \lambda^{-6} + o(\lambda^{-6})$, then the 1st-order metric coefficient converges to $4C_R(E_0^{-1})^2$. Furthermore, under the relative Cameron–Martin covariance convergence $B_\lambda = Q_0^{1/2}(I + K_\lambda)Q_0^{1/2}$ with $\|K_\lambda\| \to 0$, the limit is generated by the Cameron–Martin energy of the off-diagonal coupling vector.

Proof. Substituting $A_\lambda^{-1} = 2\lambda^2 E_0^{-1}$ into the Woodbury expansion yields:

$$S_{K,\lambda}^{-1} = 2\lambda^2 E_0^{-1} + (2\lambda^2 E_0^{-1}) R_\lambda (2\lambda^2 E_0^{-1}) + \mathcal{O}(R_\lambda^2). \tag{40}$$

Multiplying by the pull-back Jacobian λ^{-2}:

$$\lambda^{-2} S_{K,\lambda}^{-1} = 2E_0^{-1} + \lambda^{-2}(4\lambda^4 E_0^{-1} R_\lambda E_0^{-1}) + \mathcal{O}(\lambda^{-2} R_\lambda^2). \tag{41}$$

Rearranging the powers of λ in the second term gives $\lambda^{-4}(2E_0^{-1})(\lambda^6 R_\lambda)(2E_0^{-1})$. Since $R_\lambda = O(\lambda^{-6})$, the remainder is $O(\lambda^{-14})$, which becomes $\mathcal{O}(\lambda^{-8})$ relative to the expansion. The identification of C_R under the relative Cameron–Martin convergence is proved in Section C via a Fenchel-dual variational limit. □

5.4. **Displacement flatness and the 0th-order tangent cone.**

Corollary 17 (Displacement flatness)**.** *The 0th-order normal geometry is the constant metric $g^{(0)} = 2E_0^{-1}$. Consequently, all geometric objects related to its curvature vanish identically in the blow-up coordinates:*

$$\Gamma^k_{ij} = 0, \qquad R^\ell{}_{ijk} = 0, \qquad \mathrm{Scal} = 0. \tag{42}$$

The structure separates the boundary behaviors cleanly. The 0th-order term yields the constant $2E_0^{-1}$, eliminating spurious boundary curvature ($\Gamma = 0$). The singular limit preserves the canonical flat Wasserstein displacement property.

Remark 18 (Displacement flatness and the W_2 geodesic hyperplane)*.* The flatness of a Gaussian location model with fixed covariance is a priori known. The nontrivial conclusion of the unified blow-up metric theorem is that the topological clamp of the singular boundary does not manufacture spurious curvature. Instead, the limit preserves the original model's *displacement flatness*: in the W_2 Wasserstein metric, the conditional expectations respond to the conditioning value by translating along a constant L^2 displacement vector. The rescaled evidence manifold behaves as a exact geodesic hyperplane in W_2 space. This explains why the 0th-order term must be flat. (We emphasize that this is a structural geometric observation; no Wasserstein gradient-flow identity is used in the derivations).

5.5. **Ehresmann horizontal lift and the 1st-order perturbation.**

The first-order $O(\lambda^{-4})$ perturbation in Theorem 16 accurately identifies the geometry of conditional bulk shifting. We can understand this by constructing a fiber bundle over the joint Gaussian model:

(1) **Base Manifold** $\mathcal{B}$: The conditioning state space of the detector $\{x_K^*\}$.
(2) **Fiber** $\mathcal{F}$: The unobserved bulk state space μ_I.
(3) **Total Space** $\mathcal{E}$: The joint Gaussian model.

When the conditioning value x_K^* is moved, the bulk conditional distribution $\mu_{I|K}(x_K^*)$ undergoes a deterministic shift. The bulk conditional mean is:

$$m_{I|K}(x_K^*) = m_I + c_\lambda a_\lambda^{-1}(x_K^* - m_K). \tag{43}$$

Remark 19 (Fiber-bundle interpretation and the horizontal connection)*.* By considering the conditional expectation as an Ehresmann section into the target fiber, its derivative mapping defines an *Ehresmann connection*:

$$Ds_\lambda = \frac{d}{dx_K^*} m_{I|K} = c_\lambda a_\lambda^{-1}. \tag{44}$$

This derivative is exactly the *horizontal lift vector*—the lift of the base tangent vector $\partial/\partial x_K^*$ into the total space. The extracted first-order residue evaluates exactly the inner metric trace of this structural bundle response (in the B_λ^{-1} norm).

Thus, the collapse of the bulk Fisher metric $g_\lambda^{\text{bulk}} = O(\lambda^{-2}) \to 0$ has a clear geometric meaning: **as the conditioning strength $\lambda \to \infty$, the connection tends to become "purely vertical."** Movements in the base manifold increasingly fail to induce horizontal drift in the bulk fiber. The RFIM limit $\widetilde{g}_{\text{bulk}}^{\text{FP}} = \lim \lambda^2 g_\lambda^{\text{bulk}}$ quantifies the asymptotic rate of this horizontal leakage, capturing the Cameron–Martin energy of the connection.

6. Elliptic Gaussian Field Example

The numerical diagnostics in the appendix validate the unified blow-up geometry on a finite-dimensional chain. In this section we show that the abstract RFIM invariant of the 1st-order perturbation specializes to a concrete Sobolev–Dirichlet energy integral on a genuinely infinite-dimensional Gaussian field model.

6.1. Elliptic Gaussian field on $[0,1]$.

Let $\mathcal{H} = L^2(0,1)$ and consider the second-order elliptic operator

$$L = -\partial_{xx} + \kappa^2, \qquad \kappa > 0,$$

with Dirichlet boundary conditions on $[0,1]$. The operator L is self-adjoint, strictly positive, and has the eigenbasis $e_j(x) = \sqrt{2}\sin(j\pi x)$ with eigenvalues

$$\mu_j = \pi^2 j^2 + \kappa^2, \qquad j = 1, 2, \ldots$$

Define the covariance operator $Q_0 = L^{-1}$. Its eigenvalues are $q_j = (\pi^2 j^2 + \kappa^2)^{-1}$, and

$$\mathrm{Tr}(Q_0) = \sum_{j=1}^{\infty} \frac{1}{\pi^2 j^2 + \kappa^2} < \infty,$$

so Q_0 is trace-class on $L^2(0,1)$. The centered Gaussian measure $\mu_0 = \mathcal{N}(0, Q_0)$ is the equilibrium law of the Hilbert-space Ornstein–Uhlenbeck process

$$dX_t = -LX_t\,dt + dW_t,$$

where W_t is a cylindrical Wiener process on $L^2(0,1)$.

6.2. **Cameron–Martin space and energy.** The Cameron–Martin space of μ_0 is

$$\mathrm{Dom}(Q_0^{-1/2}) = \mathrm{Dom}(L^{1/2}) = H_0^1(0,1),$$

the Sobolev space of square-integrable functions with square-integrable weak derivatives vanishing at the boundary. The Cameron–Martin energy for $a \in H_0^1(0,1)$ is

$$\|a\|_{Q_0^{-1}}^2 = \|L^{1/2} a\|_{L^2}^2 = \int_0^1 |a'(x)|^2\, dx + \kappa^2 \int_0^1 |a(x)|^2\, dx. \tag{45}$$

This is the classical Dirichlet energy with mass term, naturally associated with the elliptic operator L.

6.3. **Spectral sensor conditioning.** Let $K = \{k\}$ for a fixed mode index k, and set $\mathcal{H}_K = \mathrm{span}(e_k)$, $\mathcal{H}_I = \overline{\mathrm{span}}(e_j : j \neq k)$. The "sensor coordinate" is the spectral coefficient

$$X_k = \langle X, e_k \rangle_{L^2},$$

which is a bounded linear functional on $L^2(0,1)$ and hence defines a well-posed target coordinate in the framework of Section 5.

The unperturbed target variance is $\Sigma_{0,kk} = q_k = (\pi^2 k^2 + \kappa^2)^{-1}$, and the evidence noise coefficient is $\varepsilon_0 = 2\mu_k q_k = 2$. Cross-mode coupling, which produces a nontrivial Cameron–Martin leakage vector a_{Ik}, arises from bounded off-diagonal perturbations to the drift operator. Specifically, if the full drift is $\mathcal{A} = -L + V$ where $V \in \mathcal{L}(\mathcal{H})$ is a bounded perturbation with $\langle e_j, V e_k \rangle \neq 0$ for at least one $j \neq k$, then the stationary cross-covariance $\Sigma_{0,Ik}$ is nonzero and generates a nontrivial coupling vector $a_{Ik} \in H_0^1(0,1)$.

6.4. **RFIM as Dirichlet connection energy.** Applying Theorem 16 under the relative Cameron–Martin covariance convergence, the renormalized 1st-order horizontal metric limit takes the explicit integral form:

$$\boxed{\widetilde{g}^{\mathrm{FP}}_{\mathrm{bulk}} = \|a_{Ik}\|_{Q_0^{-1}}^2 = \int_0^1 |a'_{Ik}(x)|^2\, dx + \kappa^2 \int_0^1 |a_{Ik}(x)|^2\, dx.} \tag{46}$$

This converts the abstract RFIM invariant of the tangent blow-up geometry into a recognizable PDE quantity: the energy of the coupling function a_{Ik} in the Cameron–Martin (Sobolev) norm associated with the equilibrium covariance.

In the language of fiber bundles and Ehresmann connections established in Theorem 19, (46) elevates the RFIM limit from a simple scalar into an *infinite-dimensional connection energy functional.* The L^2-energy of the connection 1-form is precisely the Cameron–Martin/Dirichlet norm. A smooth coupling function $a_{Ik} \in H_0^1$ with small Dirichlet energy produces a weak horizontal lift perturbation, while a rough coupling produces a strong geometric perturbation on the bundle.

7. Discussion

We have demonstrated that singular Gaussian conditioning in a separable Hilbert space admits an inverse-free Schur normal form, where the rescaled Fisher tensor decomposes into a canonical leading flat layer and a Cameron–Martin residue. By bypassing the unbounded covariance inverse operator through a variational projection residual, the structural properties of the singular boundary are preserved without the need to introduce artificial noise or ad hoc subtraction of infinite quantities.

Several natural directions for future research remain:

(i) **Non-Gaussian Conditional Finite-Part Functionals:** The projection residual scaling $S_{k,\lambda} = \varepsilon_0/(2\lambda^2) - R_\lambda$ holds for any linear stationary system with finite second moments. Extending the Gaussian finite-part functional analysis to non-Gaussian stationary processes (e.g., driven by Lévy or Laplace noise) requires a systematic generalization of the Radon disintegration and Feldman–Hájek theory.

(ii) **Transport Relaxation Dynamics:** While the tangent blow-up coordinate system provides a static, local description of the singular Fisher geometry on the boundary, analyzing the dynamic relaxation of the bulk covariance after singular conditioning requires the transport geometry of the Bures–Wasserstein metric. We leave the JKO gradient flow [12] on the Wasserstein fibers to subsequent work.

8. Conclusion

We have established a Hilbert-space framework for singular hyperplane conditioning on separable Hilbert spaces. The projection residual scaling holds for the stated class of linear finite-second-moment stationary models, while the Gaussian renormalized finite-part functional $\mathfrak{J}_\lambda$ converges to a finite $O(1)$ constant (and is exactly zero under decoupled centering). The numerical diagnostics and Galerkin verification in Section D illustrate truncation behavior, exact variance normalization, and Gaussian finite-part asymptotics.

Declaration of Generative AI and AI-Assisted Technologies in the Manuscript Preparation Process

During the preparation of this work, the author(s) used Google DeepMind's Antigravity assistant in order to merge the LaTeX manuscripts, clean up cross-referencing tags, format mathematical notation, and edit the draft for style consistency. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

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Appendix A. Exact Block and Variational Schur Analysis

This appendix derives the linear invariants from the block Lyapunov equation and proves the strong convergence of Galerkin truncations. We do not use Laurent series or covariance inverses.

A.1. Exact Target Covariance Block.

A.1. Exact Target Covariance Block. Write $\mathcal{H} = \mathcal{H}_k \oplus \mathcal{H}_I$ with $\mathcal{H}_k = \mathbb{R}\phi_k$. The incoming-coordinate cut and the calibrated noise scale give the block operators:

$$\mathcal{A}^{(\lambda)} = \begin{pmatrix} -\lambda & 0 \\ a_{Ik} & \mathcal{A}_{II} \end{pmatrix}, \qquad \mathcal{D}^{(\lambda)} = \begin{pmatrix} \varepsilon_0/\lambda & 0 \\ 0 & D_{II} \end{pmatrix}. \tag{47}$$

The block $\Sigma_{\lambda,Ik} \in \mathcal{H}_I$ denotes the vector representation of the target-to-bulk cross-covariance. The kk block of the Lyapunov equation is the scalar equation:

$$-2\lambda\Sigma_{\lambda,kk} + \frac{\varepsilon_0}{\lambda} = 0 \quad \Longrightarrow \quad \Sigma_{\lambda,kk} = \frac{\varepsilon_0}{2\lambda^2}. \tag{48}$$

This equality holds exactly for all $\lambda > 0$.

A.2. Exact Cross-Covariance Resolvent Formula. The Ik block of the Lyapunov equation reduces to:

$$(\mathcal{A}_{II} - \lambda I)\Sigma_{\lambda,Ik} + a_{Ik}\Sigma_{\lambda,kk} = 0. \tag{49}$$

This yields the exact cross-covariance resolvent formula:

$$\Sigma_{\lambda,Ik} = \frac{\varepsilon_0}{2\lambda^2}(\lambda I - \mathcal{A}_{II})^{-1} a_{Ik}. \tag{50}$$

A.3. Cameron–Martin Energy Bound. Let $Q_0 := \Sigma_{II}^{(0)}$ be the background bulk covariance. The Cameron–Martin space $H_{Q_0} := \mathrm{Range}(Q_0^{1/2})$ has energy norm $|u|_{Q_0} := \|Q_0^{-1/2}u\|_{\mathcal{H}_I}$. Under Assumption 6, the restricted resolvent satisfies the energy bound:

$$|(\lambda I - \mathcal{A}_{II})^{-1} u|_{Q_0} \le \frac{M}{\lambda + \omega}|u|_{Q_0} \tag{51}$$

for all $u \in H_{Q_0}$. It follows from (50) that the cross-covariance has finite energy norm and decays as:

$$|\Sigma_{\lambda,Ik}|_{Q_0} \le \frac{\varepsilon_0 M}{2\lambda^2(\lambda+\omega)}|a_{Ik}|_{Q_0} = O(\lambda^{-3}). \tag{52}$$

A.4. Variational Schur Residual Proof. We define the variational Schur subtraction R_λ:

$$R_\lambda = \sup_{\langle u, \Sigma_{\lambda,II}u\rangle > 0} \frac{1}{\langle u, \Sigma_{\lambda,II}u\rangle}|\langle u, \Sigma_{\lambda,Ik}\rangle|^2. \tag{53}$$

Because the target and bulk noise channels are independent, the bulk covariance dominates the background covariance ($\Sigma_{\lambda,II} = Q_0 + \mathrm{Cov}(Y_I) \succeq Q_0$). We then have:

$$0 \le R_\lambda \le |\Sigma_{\lambda,Ik}|^2_{Q_0} \le \frac{\varepsilon_0^2 M^2 |a_{Ik}|^2_{Q_0}}{4\lambda^4(\lambda+\omega)^2} = O(\lambda^{-6}). \tag{54}$$

The projection residual satisfies $S_{k,\lambda} = \Sigma_{\lambda,kk} - R_\lambda$, so substituting the exact target block and variational remainder bound yields:

$$S_{k,\lambda} = \frac{\varepsilon_0}{2\lambda^2} - O(\lambda^{-6}), \qquad m_2^{\mathrm{Schur}} = \frac{\varepsilon_0}{2}, \tag{55}$$

which implies the inverse scaling:

$$S_{k,\lambda}^{-1} = \frac{2}{\varepsilon_0}\lambda^2 + O(\lambda^{-2}). \tag{56}$$

A.5. **Galerkin Truncation Convergence.** Here we prove Proposition 10. The proof has two layers: a dynamic Bochner-integral argument that gives trace-norm convergence of the truncated covariance operators, followed by a spectral Schur approximation that yields monotone convergence and an explicit rate.

Proposition 20 (Exact Dynamic Galerkin Convergence Support)**.** *Let $\Pi_n : \mathcal{H} \to \mathcal{H}_n$ be finite-rank orthogonal Galerkin projections such that*

$$\Pi_n \phi_k = \phi_k, \qquad \Pi_n \mathcal{H}_I \subset \mathcal{H}_I, \qquad \Pi_n \to I \tag{57}$$

strongly on $\mathcal{H}$. For each $\lambda > 0$, define the truncated regularized dynamics by

$$\mathcal{A}_n^{(\lambda)} := \Pi_n \mathcal{A}^{(\lambda)}|_{\mathcal{H}_n}, \qquad \mathcal{D}_n^{(\lambda)} := \Pi_n \mathcal{D}^{(\lambda)} \Pi_n. \tag{58}$$

Assume that $e^{t\mathcal{A}_n^{(\lambda)}} x \to e^{t\mathcal{A}^{(\lambda)}} x$ for every $x \in \mathcal{H}$, uniformly for t in compact intervals, and that the semigroups are uniformly exponentially stable:

$$\|e^{t\mathcal{A}_n^{(\lambda)}}\| \le M e^{-\beta t}, \qquad \|e^{t\mathcal{A}^{(\lambda)}}\| \le M e^{-\beta t}, \tag{59}$$

with $M \ge 1$, $\beta > 0$ independent of n. Assume also

$$\mathcal{D}_n^{(\lambda)} \to \mathcal{D}^{(\lambda)} \quad \text{in } \mathcal{S}_1(\mathcal{H}). \tag{60}$$

Then the Galerkin stationary covariance operators

$$\Sigma_{n,\lambda} = \int_0^\infty e^{t\mathcal{A}_n^{(\lambda)}} \mathcal{D}_n^{(\lambda)} e^{t\mathcal{A}_n^{(\lambda)*}}\, dt \tag{61}$$

converge in trace norm:

$$\Sigma_{n,\lambda} \to \Sigma_\lambda \quad \text{in } \mathcal{S}_1(\mathcal{H}). \tag{62}$$

Moreover, because the Galerkin scheme preserves the target-bulk block decomposition, one has the exact finite-n target calibration

$$\Sigma_{n,\lambda,kk} = \frac{\varepsilon_0}{2\lambda^2} \tag{63}$$

for every $n \geq k$. If the off-diagonal drift coupling satisfies the uniform finite-energy condition

$$a_{Ik}^{(n)} \in H_{Q_0^{(n)}}, \qquad \sup_n |a_{Ik}^{(n)}|_{Q_0^{(n)}} < \infty, \tag{64}$$

where $Q_0^{(n)} := \Sigma_{n,II}^{(0)}$ is the truncated background covariance, then the truncated Schur residual satisfies

$$S_{k,\lambda}^{(n)} = \frac{\varepsilon_0}{2\lambda^2} - R_{n,\lambda}, \qquad 0 \leq R_{n,\lambda} \leq C\lambda^{-6}. \tag{65}$$

Consequently,

$$m_2^{\mathrm{Schur},(n)} := \lim_{\lambda\to\infty} \lambda^2 S_{k,\lambda}^{(n)} = \frac{\varepsilon_0}{2} \tag{66}$$

for every admissible Galerkin truncation $n \geq k$.

Proof. Write the difference of covariances as the Bochner integral

$$\Sigma_{n,\lambda} - \Sigma_\lambda = \int_0^\infty \left(e^{t\mathcal{A}_n^{(\lambda)}} \mathcal{D}_n^{(\lambda)} e^{t\mathcal{A}_n^{(\lambda)*}} - e^{t\mathcal{A}^{(\lambda)}} \mathcal{D}^{(\lambda)} e^{t\mathcal{A}^{(\lambda)*}} \right) dt. \tag{67}$$

Split the integrand in the trace norm:

$$\begin{aligned} \left\| e^{t\mathcal{A}_n} \mathcal{D}_n e^{t\mathcal{A}_n^*} - e^{t\mathcal{A}} \mathcal{D} e^{t\mathcal{A}^*} \right\|_{\mathcal{S}_1} \leq{} & \left\| e^{t\mathcal{A}_n} (\mathcal{D}_n - \mathcal{D}) e^{t\mathcal{A}_n^*} \right\|_{\mathcal{S}_1} \\ & + \left\| e^{t\mathcal{A}_n} \mathcal{D} e^{t\mathcal{A}_n^*} - e^{t\mathcal{A}} \mathcal{D} e^{t\mathcal{A}^*} \right\|_{\mathcal{S}_1}. \end{aligned} \tag{68}$$

(We omit the superscript (λ) for readability.)

The first term is bounded by $M^2 e^{-2\beta t} \|\mathcal{D}_n - \mathcal{D}\|_{\mathcal{S}_1}$, which tends to zero by assumption. For the second term, we use the two-sided ideal convergence property of $\mathcal{S}_1(\mathcal{H})$ (Grümm's theorem; see [18]): if $L_n \to L$ strongly, $R_n^* \to R^*$ strongly, the two sequences are uniformly bounded, and $K \in \mathcal{S}_1(\mathcal{H})$, then $L_n K R_n \to LKR$ in $\mathcal{S}_1(\mathcal{H})$. Applying this with $L_n = e^{t\mathcal{A}_n}$, $R_n = e^{t\mathcal{A}_n^*}$, $L = e^{t\mathcal{A}}$, $R = e^{t\mathcal{A}^*}$, and $K = \mathcal{D}$, the second term converges to zero for each t. Crucially, $R_n^* = e^{t\mathcal{A}_n} \to e^{t\mathcal{A}} = R^*$ is already covered by the same Trotter–Kato strong convergence assumption; no independent strong convergence of the adjoint semigroups is required. The exponential stability gives the integrable dominating bound

$$\left\| e^{t\mathcal{A}_n} \mathcal{D}_n e^{t\mathcal{A}_n^*} - e^{t\mathcal{A}} \mathcal{D} e^{t\mathcal{A}^*} \right\|_{\mathcal{S}_1} \leq M^2 e^{-2\beta t} \big(\|\mathcal{D}_n\|_{\mathcal{S}_1} + \|\mathcal{D}\|_{\mathcal{S}_1} \big) \leq M^2 C_{\mathcal{D}}\, e^{-2\beta t}, \tag{69}$$

where $C_{\mathcal{D}} := \sup_n \|\mathcal{D}_n\|_{\mathcal{S}_1} + \|\mathcal{D}\|_{\mathcal{S}_1} < \infty$ (the supremum is finite because $\mathcal{D}_n \to \mathcal{D}$ in trace norm). The dominated-convergence theorem for Bochner integrals then yields

$$\|\Sigma_{n,\lambda} - \Sigma_\lambda\|_{\mathcal{S}_1} \to 0. \tag{70}$$

Because Π_n preserves $\mathcal{H}_k \oplus \mathcal{H}_I$, the truncated blocks remain

$$\mathcal{A}_n^{(\lambda)} = \begin{pmatrix} -\lambda & 0 \\ a_{Ik}^{(n)} & \mathcal{A}_{II}^{(n)} \end{pmatrix}, \qquad \mathcal{D}_n^{(\lambda)} = \begin{pmatrix} \varepsilon_0/\lambda & 0 \\ 0 & \mathcal{D}_{II}^{(n)} \end{pmatrix}. \tag{71}$$

Thus the (kk)-block of the truncated Lyapunov equation is exactly

$$-2\lambda\,\Sigma_{n,\lambda,kk} + \frac{\varepsilon_0}{\lambda} = 0, \tag{72}$$

hence $\Sigma_{n,\lambda,kk} = \varepsilon_0/(2\lambda^2)$. The (Ik)-block gives

$$(\mathcal{A}_{II}^{(n)} - \lambda I)\Sigma_{n,\lambda,Ik} + a_{Ik}^{(n)}\Sigma_{n,\lambda,kk} = 0, \tag{73}$$

so

$$\Sigma_{n,\lambda,Ik} = \frac{\varepsilon_0}{2\lambda^2}(\lambda I - \mathcal{A}_{II}^{(n)})^{-1} a_{Ik}^{(n)}. \tag{74}$$

Uniform Cameron–Martin stability (the finite-energy condition on $a_{Ik}^{(n)}$) gives $|\Sigma_{n,\lambda,Ik}|_{Q_0^{(n)}} = O(\lambda^{-3})$ uniformly in n. Therefore the variational Schur subtraction satisfies

$$0 \le R_{n,\lambda} \le |\Sigma_{n,\lambda,Ik}|^2_{Q_0^{(n)}} = O(\lambda^{-6}), \tag{75}$$

and hence

$$S_{k,\lambda}^{(n)} = \frac{\varepsilon_0}{2\lambda^2} - O(\lambda^{-6}). \tag{76}$$

Multiplying by λ^2 and letting $\lambda \to \infty$ gives $m_2^{\mathrm{Schur},(n)} = \varepsilon_0/2$. □

Lemma 21 (Spectral Schur Compression Support)**.** *Fix $\lambda > 0$ and let $C_\lambda := \Sigma_{\lambda,II}$ denote the bulk covariance operator on $\mathcal{H}_I$. Let $\{(\sigma_j, e_j)\}_{j=1}^{\infty}$ be the eigenpairs of C_λ with $\sigma_1 \ge \sigma_2 \ge \cdots > 0$. Define the spectral compression projectors*

$$P_n^\lambda := \sum_{j=1}^{n} e_j \otimes e_j, \qquad n \ge 1. \tag{77}$$

Under Assumption 6 (Cameron–Martin stability), the leakage vector satisfies $\Sigma_{\lambda,Ik} \in \mathrm{Range}(C_\lambda^{1/2})$, so the spectral Schur compression

$$R_\lambda^{[n]} := \sum_{j=1}^{n} \frac{|\langle \Sigma_{\lambda,Ik}, e_j\rangle|^2}{\sigma_j} \tag{78}$$

is finite for every n and monotone increasing:

$$R_\lambda^{[n]} \uparrow R_\lambda := \|\Sigma_{\lambda,Ik}\|^2_{CM} \quad \text{as } n \to \infty. \tag{79}$$

Consequently the compressed Schur residuals satisfy

$$S_{k,\lambda}^{[n]} := \frac{\varepsilon_0}{2\lambda^2} - R_\lambda^{[n]} \quad \searrow \quad S_{k,\lambda} := \frac{\varepsilon_0}{2\lambda^2} - R_\lambda \quad \text{as } n \to \infty. \tag{80}$$

If, moreover, the weighted Cameron–Martin tail condition

$$\sum_{j\ge 1} j^{\,\eta-1} \frac{|\langle \Sigma_{\lambda,Ik}, e_j\rangle|^2}{\sigma_j} \le C\lambda^{-6}, \qquad \eta > 1, \tag{81}$$

holds, then the tail is explicitly bounded by

$$S_{k,\lambda}^{[n]} - S_{k,\lambda} = \sum_{j=n+1}^{\infty} \frac{|\langle \Sigma_{\lambda,Ik}, e_j \rangle|^2}{\sigma_j} \le C\, n^{1-\eta} \lambda^{-6}. \tag{82}$$

Proof. By Assumption 6, $\Sigma_{\lambda,Ik} \in H_{Q_0} \subseteq \mathrm{Range}(C_\lambda^{1/2})$. Hence $C_\lambda^{-1/2}\Sigma_{\lambda,Ik} \in \mathcal{H}_I$ and

$$R_\lambda^{[n]} = \sum_{j=1}^{n} \frac{|\langle \Sigma_{\lambda,Ik}, e_j \rangle|^2}{\sigma_j} = \|P_n^\lambda C_\lambda^{-1/2} \Sigma_{\lambda,Ik}\|_{\mathcal{H}_I}^2 \le \|C_\lambda^{-1/2} \Sigma_{\lambda,Ik}\|_{\mathcal{H}_I}^2 = R_\lambda. \tag{83}$$

Since $P_n^\lambda \to I$ strongly and the terms are non-negative, $R_\lambda^{[n]} \uparrow R_\lambda$. The exact target block (48) gives $\Sigma_{\lambda,kk} = \varepsilon_0/(2\lambda^2)$, so $S_{k,\lambda}^{[n]} = \varepsilon_0/(2\lambda^2) - R_\lambda^{[n]} \searrow S_{k,\lambda}$. Using (52), $R_\lambda = O(\lambda^{-6})$, whence $S_{k,\lambda} = \varepsilon_0/(2\lambda^2) - O(\lambda^{-6})$ and $m_2^{\mathrm{Schur}} = \varepsilon_0/2$.

Under the weighted tail condition, the remainder after n terms satisfies

$$\sum_{j=n+1}^{\infty} \frac{|\langle \Sigma_{\lambda,Ik}, e_j \rangle|^2}{\sigma_j} \le n^{1-\eta} \sum_{j=n+1}^{\infty} j^{\,\eta-1} \frac{|\langle \Sigma_{\lambda,Ik}, e_j \rangle|^2}{\sigma_j} \le C\, n^{1-\eta} \lambda^{-6}, \tag{84}$$

which gives the asserted rate. □

Appendix B. Gaussian Conditional Finite-Part Analysis

This appendix derives the Radon Gaussian conditioning, measurable conditional predictor, bulk relative entropy bounds, and the finite-part regularization theorem via Dirac mollification.

B.1. Radon Gaussian Setup and Pointwise Disintegration.

Let $\mu_\lambda = \mathcal{N}(m_\lambda, \Sigma_\lambda)$ be a Radon Gaussian measure on the separable Hilbert space $\mathcal{H} = \mathcal{H}_k \oplus \mathcal{H}_I$. The trace-class property of Σ_λ ensures its existence as a Radon measure [4]. Let $\pi_k : \mathcal{H} \to \mathbb{R}$ be the projection $\pi_k(X) = \langle X, \phi_k \rangle = X_k$.

Lemma 22 (Continuous Disintegration and Exact Target Zeros)**.** *For each $t \in \mathbb{R}$, there exists a unique weakly continuous family of Gaussian Radon measures $\mu_\lambda^t = \mathcal{N}(m_\lambda^t, \Sigma_\lambda^t)$ supported on the hyperplane $X_k = t$, where:*

$$m_\lambda^t := \begin{pmatrix} t \\ m_{\lambda,I} + \Sigma_{\lambda,Ik}\Sigma_{\lambda,kk}^{-1}(t - m_{\lambda,k}) \end{pmatrix}, \qquad \Sigma_\lambda^t := \begin{pmatrix} 0 & 0 \\ 0 & \Sigma_{\lambda,II} - \Sigma_{\lambda,Ik}\Sigma_{\lambda,kk}^{-1}\Sigma_{\lambda,kI} \end{pmatrix}. \tag{85}$$

Under the conditioned law $\mu_{\mathrm{obs}} := \mu_\lambda^{x_k^*}$, the target coordinate satisfies the exact target zeros:

$$\mathbb{E}_{\mathrm{obs}}[X_k] - x_k^* = 0, \quad \mathrm{Var}_{\mathrm{obs}}(X_k) = 0. \tag{86}$$

Proof. Since $\mathcal{H}$ is Polish, regular conditional probabilities exist. The stated Gaussian family is the continuous disintegration along the scalar observation map; its pointwise continuous version is unique [13]. The continuity of $t \mapsto m_\lambda^t$ in $\mathcal{H}$ and constancy of Σ_λ^t yield weak continuity of this family (see also [4]). Evaluating at $t = x_k^*$ gives $\mu_{\text{obs}} = \mu_\lambda^{x_k^*}$, which has target component fixed at x_k^* almost surely. □

B.2. **Measurable Conditional Predictor.** Let $C_\lambda \coloneqq \Sigma_{\lambda,II}$ be the bulk covariance on $\mathcal{H}_I$. Since C_λ is trace-class, it lacks a bounded global inverse. We define the bulk Cameron–Martin space $H_{C_\lambda} \coloneqq \mathrm{Range}(C_\lambda^{1/2})$.

Lemma 23 (Measurable Conditional Predictor)**.** *The cross-covariance vector satisfies $\Sigma_{\lambda,Ik} \in H_{C_\lambda}$ with energy norm:*

$$\|\Sigma_{\lambda,Ik}\|_{H_{C_\lambda}} \le |\Sigma_{\lambda,Ik}|_{Q_0} = O(\lambda^{-3}). \tag{87}$$

Furthermore, the conditional mean predictor $\mu_{k|I}(X_I) \coloneqq \mathbb{E}_{\mu_\lambda}[X_k \mid X_I]$ is represented by the measurable Wiener functional:

$$\mu_{k|I}(X_I) = m_{\lambda,k} + W_{\Sigma_{\lambda,Ik}}(X_I - m_{\lambda,I}), \tag{88}$$

where $W_{\Sigma_{\lambda,Ik}}$ is the $L^2(\mu_{\lambda,I})$-limit of the continuous linear functionals on $\mathcal{H}_I$. For all sufficiently large λ, the conditional variance is strictly positive:

$$\mathrm{Var}(X_k \mid X_I) = \Sigma_{\lambda,kk} - \|\Sigma_{\lambda,Ik}\|^2_{H_{C_\lambda}} = S_{k,\lambda} > 0. \tag{89}$$

Proof. Since $\Sigma_{\lambda,II} \succeq Q_0$, Douglas range inclusion [7] yields $\mathrm{Range}(Q_0^{1/2}) \subseteq \mathrm{Range}(\Sigma_{\lambda,II}^{1/2}) = H_{C_\lambda}$ with the contraction inequality

$$\|\Sigma_{\lambda,II}^{-1/2} v\|_{\mathcal{H}_I} \le \|Q_0^{-1/2} v\|_{\mathcal{H}_I} \quad \text{for all } v \in \mathrm{Range}(Q_0^{1/2}).$$

Hence, $\Sigma_{\lambda,Ik} \in H_{C_\lambda}$ and the energy norm bound follows directly from (52). The range inclusion $\Sigma_{\lambda,Ik} \in \mathrm{Range}(\Sigma_{\lambda,II}^{1/2})$ guarantees that the Wiener functional $W_{\Sigma_{\lambda,Ik}}$ is well-defined as a measurable function on $(\mathcal{H}_I, \mu_{\lambda,I})$ [4]. The conditional variance is the projection residual. By Theorem 9,

$$S_{k,\lambda} = \frac{\varepsilon_0}{2\lambda^2} - O(\lambda^{-6}) > 0$$

for all sufficiently large λ. □

B.3. **Bulk Relative-Entropy Estimate.**

Lemma 24 (Feldman–Hájek Equivalence and Entropy Decay)**.** *There exists $\lambda_0 > 0$ such that, for every $\lambda \ge \lambda_0$, the bulk marginals $\mu_{\text{obs},I}$ and $\mu_{\lambda,I}$ are equivalent Gaussian Radon measures on $\mathcal{H}_I$, with relative entropy:*

$$\begin{aligned} 2D_{\mathrm{KL}}(\mu_{\text{obs},I} \parallel \mu_{\lambda,I}) = {} & \frac{(x_k^* - m_{\lambda,k})^2}{\Sigma_{\lambda,kk}} \rho_\lambda \\ & - \log(1 - \rho_\lambda) - \rho_\lambda, \end{aligned} \tag{90}$$

where the eigenvalue $\rho_\lambda := \|\Sigma_{\lambda,Ik}\|^2_{H_{C_\lambda}}/\Sigma_{\lambda,kk} = 1 - S_{k,\lambda}/\Sigma_{\lambda,kk} < 1$. *In particular, we have* $\rho_\lambda = O(\lambda^{-4})$ *and:*

$$D_{\mathrm{KL}}(\mu_{\mathrm{obs},I} \parallel \mu_{\lambda,I}) = O(\lambda^{-4}) \to 0 \quad \textit{as } \lambda \to \infty. \tag{91}$$

Proof. Under disintegration, $\mu_{\mathrm{obs},I}$ is a Gaussian Radon law on $\mathcal{H}_I$ with mean $m_{\mathrm{obs},I} = m_{\lambda,I} + \Sigma_{\lambda,Ik}\Sigma^{-1}_{\lambda,kk}(x^*_k - m_{\lambda,k})$ and covariance $\Sigma_{\mathrm{obs},II} = C_\lambda - \Sigma_{\lambda,Ik}\Sigma^{-1}_{\lambda,kk}\Sigma_{\lambda,kI}$. The covariance operators differ by the rank-one operator $T_\lambda = \Sigma_{\lambda,Ik}\Sigma^{-1}_{\lambda,kk}\Sigma_{\lambda,kI}$. Since $\Sigma_{\lambda,Ik} \in H_{C_\lambda}$, the relative covariance operator is a rank-one perturbation of the identity with its only non-unit eigenvalue equal to $1-\rho_\lambda$. By Lemma 23 and the exact target covariance, $\rho_\lambda = O(\lambda^{-4})$. Choose λ_0 so that $\rho_\lambda < 1$ for $\lambda \geq \lambda_0$. The Feldman–Hájek theorem [8, 4] then guarantees equivalence of the measures, and the relative entropy is:

$$\begin{aligned} 2D_{\mathrm{KL}}(\mu_{\mathrm{obs},I} \parallel \mu_{\lambda,I}) = \|C_\lambda^{-1/2}(m_{\mathrm{obs},I} - m_{\lambda,I})\|^2_{\mathcal{H}_I} \\ - \log(1-\rho_\lambda) - \rho_\lambda. \end{aligned}$$

The mean-shift identity in the lemma follows from

$$m_{\mathrm{obs},I} - m_{\lambda,I} = \Sigma_{\lambda,Ik}\Sigma^{-1}_{\lambda,kk}(x^*_k - m_{\lambda,k}).$$

Assumption 11 gives $m_{\lambda,k} - x^*_k = O(\lambda^{-1})$, so that term is $O(\lambda^{-4})$. Finally,

$$-\log(1-\rho_\lambda) - \rho_\lambda = O(\rho_\lambda^2) = O(\lambda^{-8}),$$

which proves the asserted entropy decay. □

B.4. **Mollified Divergence and Finite-Part Regularization Proof.** To regularize the singular joint KL divergence, define a law with bulk marginal $\mu_{\mathrm{obs},I}$ and an independent mollified target conditional:

$$\nu_{\lambda,\delta} := \mu_{\mathrm{obs},I} \otimes \mathcal{N}(x^*_k, \delta^2) \tag{92}$$

for $\delta > 0$, identifying $\mathcal{H}$ with $\mathcal{H}_I \times \mathcal{H}_k$ in this display.

Proposition 25 (Mollification Limit)**.** *For* $\lambda \geq \lambda_0$ *as in Lemma 24, define*

$$C_{\lambda,\delta} := \frac{1}{2}\left(\log\frac{S_{k,\lambda}}{\delta^2} - 1\right), \tag{93}$$

Then

$$\lim_{\delta\downarrow 0}\left[D_{\mathrm{KL}}(\nu_{\lambda,\delta} \parallel \mu_\lambda) - C_{\lambda,\delta}\right] = \mathfrak{J}_\lambda(\mu_{\mathrm{obs}};\mu_\lambda). \tag{94}$$

Proof. Put $r_\lambda(X) := X_k - \mu_{k|I}(X_I)$. The entropy chain rule and the scalar Gaussian formula give

$$\begin{aligned} D_{\mathrm{KL}}(\nu_{\lambda,\delta} \parallel \mu_\lambda) = D_{\mathrm{KL}}(\mu_{\mathrm{obs},I} \parallel \mu_{\lambda,I}) + C_{\lambda,\delta} \\ + \frac{\delta^2}{2S_{k,\lambda}} + \frac{1}{2S_{k,\lambda}}\mathbb{E}_{\mu_{\mathrm{obs}}}[r_\lambda(X)^2]. \end{aligned} \tag{95}$$

Here $X_k = x^*_k$ under μ_{obs}. Since $S_{k,\lambda} > 0$ for $\lambda \geq \lambda_0$, the term $\delta^2/(2S_{k,\lambda})$ vanishes as $\delta \downarrow 0$. The remaining terms equal Definition 12. □

Proof of Theorem 13. Let $\Delta_\lambda := x_k^* - m_{\lambda,k}$ and $Z_\lambda := W_{\Sigma_{\lambda,Ik}}(X_I - m_{\lambda,I})$. The conditional Gaussian formulas give

$$\mathbb{E}_{\text{obs}}[Z_\lambda] = \rho_\lambda \Delta_\lambda, \tag{96}$$

$$\mathrm{Var}_{\text{obs}}(Z_\lambda) = \Sigma_{\lambda,kk}\rho_\lambda(1-\rho_\lambda). \tag{97}$$

Using $X_k = x_k^*$ under μ_{obs} and $S_{k,\lambda} = \Sigma_{\lambda,kk}(1-\rho_\lambda)$, we obtain

$$\begin{aligned} &\frac{1}{2S_{k,\lambda}}\mathbb{E}_{\mu_{\text{obs}}}[(X_k - \mu_{k|I}(X_I))^2] \\ &= \frac{\rho_\lambda}{2} + \frac{\Delta_\lambda^2}{2\Sigma_{\lambda,kk}}(1-\rho_\lambda). \end{aligned} \tag{98}$$

Here

$$\frac{\Delta_\lambda^2}{2\Sigma_{\lambda,kk}} = \frac{b_k^2}{\varepsilon_0}, \qquad \rho_\lambda = O(\lambda^{-4}).$$

The conditional term is therefore $b_k^2/\varepsilon_0 + O(\lambda^{-4})$. Adding Lemma 24 proves

$$\mathfrak{J}_\lambda(\mu_{\text{obs}};\mu_\lambda) = \frac{b_k^2}{\varepsilon_0} + O(\lambda^{-4}),$$

and hence the stated limit. □

B.5. **Exact Decoupled-Centered Consequence.** If the target coordinate is dynamically decoupled ($a_{Ik} = 0$), then $\Sigma_{\lambda,Ik} = 0$, $S_{k,\lambda} = \Sigma_{\lambda,kk} > 0$, and the bulk marginals coincide for every $\lambda > 0$. If the evidence is also centered ($x_k^* = m_{\lambda,k}$), both terms in Definition 12 vanish. Hence $\mathfrak{J}_\lambda(\mu_{\text{obs}};\mu_\lambda) = 0$ exactly for every $\lambda > 0$.

Appendix C. Proofs of the Unified Blow-Up Geometry

This section provides the formal proofs for the identification of the 1st-order RFIM limit established in Theorem 16, utilizing a Fenchel-dual variational formulation.

C.1. **Imported Schur invariant.** All results in this section use the Schur residual scaling established in Section 3. We construct the geometric objects that arise naturally from the limit.

Lemma 26 (Schur Projection Summary)**.** *By Theorem 9 and Theorem 10, the regularized-conditioning Schur residual is established in two layers. The* dynamic *layer proves trace-norm convergence of the Galerkin covariance operators and gives the exact finite-n calibration*

$$S_{k,\lambda}^{(n)} = \frac{\varepsilon_0}{2\lambda^2} - R_{n,\lambda}, \qquad 0 \le R_{n,\lambda} \le C\lambda^{-6}, \quad \forall\, n \ge k. \tag{99}$$

The spectral *layer (Theorem 21) represents the subtraction as a monotone series in the eigenbasis of $\Sigma_{\lambda,II}$, yielding $S_{k,\lambda}^{(n)} \searrow S_{k,\lambda}$. Consequently,*

$$S_{K,\lambda} = \frac{E_0}{2\lambda^2} - O(\lambda^{-6}), \tag{100}$$

for the target block covariance matrix E_0.

C.2. **Revised Proof via Fenchel-Dual Variational Limit.** We now prove the identification of the horizontal metric limit C_R in Theorem 16 under the relative Cameron–Martin factorization. Assume:

(i) (Relative Cameron–Martin covariance convergence.) There exists a bounded self-adjoint operator K_λ on $\mathcal{H}_I$ such that

$$B_\lambda = Q_0^{1/2}(I + K_\lambda)Q_0^{1/2}, \qquad \|K_\lambda\|_{\mathcal{L}(\mathcal{H}_I)} \to 0,$$

with $I + K_\lambda \geq cI$ for some $c > 0$ and all $\lambda \geq \lambda_0$.

(ii) $a_{Ik} \in \mathrm{Dom}(Q_0^{-1/2})$.

(iii) $\|\lambda^3 Q_0^{-1/2} c_\lambda - (\varepsilon_0/2)\, Q_0^{-1/2} a_{Ik}\| \to 0$.

Evaluating unbounded inverse fraction powers analytically (e.g., $B_\lambda^{-1/2}$) presents immense topological risks in infinite-dimensional spaces. We instead employ the inverse-free Fenchel-variational form naturally native to the Schur remainder:

$$R_\lambda = \sup_{u \in \mathcal{H}_I} \{2\langle c_\lambda, u\rangle - \langle u, B_\lambda u\rangle\}. \tag{101}$$

To regularize the variational form under the hypothesis $B_\lambda = Q_0^{1/2}(I + K_\lambda)Q_0^{1/2}$, we substitute $v = Q_0^{1/2}u$. Since Q_0 is strictly positive and trace-class, the range $Q_0^{1/2}(\mathcal{H}_I)$ is dense in $\mathcal{H}_I$. Setting $y_\lambda := Q_0^{-1/2}c_\lambda$:

$$\begin{aligned} R_\lambda &= \sup_{u \in \mathcal{H}_I} \left\{ 2\langle Q_0^{1/2} y_\lambda, u\rangle - \langle u, Q_0^{1/2}(I + K_\lambda)Q_0^{1/2}u\rangle \right\} \\ &= \sup_{v \text{ (dense)}} \{2\langle y_\lambda, v\rangle - \langle v, (I + K_\lambda)v\rangle\}. \end{aligned} \tag{102}$$

The substitution reduces the unbounded-inverse problem to a standard concave quadratic optimization. Strict convexity gives the maximizer $v^* = (I + K_\lambda)^{-1}y_\lambda$, so:

$$R_\lambda = \langle y_\lambda, (I + K_\lambda)^{-1}y_\lambda\rangle. \tag{103}$$

By assumption (iii), $\lambda^3 y_\lambda \to (\varepsilon_0/2)\, Q_0^{-1/2}a_{Ik}$. Since $\|K_\lambda\| \to 0$, we have $(I + K_\lambda)^{-1} \to I$ in operator norm, and the inner product converges.

Passing to the limit:

$$\lambda^6 R_\lambda = \lambda^6\langle y_\lambda, (I + K_\lambda)^{-1}y_\lambda\rangle \to \frac{\varepsilon_0^2}{4}\langle Q_0^{-1/2}a_{Ik}, Q_0^{-1/2}a_{Ik}\rangle = \frac{\varepsilon_0^2}{4}\|a_{Ik}\|^2_{Q_0^{-1}}. \tag{104}$$

Thus, the asymptotic constant of the Schur residual is exactly

$$C_R = \frac{\varepsilon_0^2}{4}\|a_{Ik}\|^2_{Q_0^{-1}}. \tag{105}$$

Substituting this into the 1st-order metric coefficient term from Theorem 16 yields:

$$4C_R(E_0^{-1})^2 = 4\left(\frac{\varepsilon_0^2}{4}\|a_{Ik}\|^2_{Q_0^{-1}}\right)(E_0^{-1})^2 = \|a_{Ik}\|^2_{Q_0^{-1}}, \tag{106}$$

where we have specialized to the single-target block $\varepsilon_0 = E_0$. This completely identifies the 1st-order horizontal metric limit. □

Appendix D. Numerical Diagnostics and Galerkin Verification

This appendix presents arbitrary-precision numerical validations of the theoretical geometric structures and Galerkin convergence rates, computed in 256-bit `BigFloat` to isolate the structural invariants from floating-point collapse.

D.1. **Truncation Convergence.** We illustrate the Galerkin convergence of Proposition 10 against a finite reference on an admissible model. The Hilbert space is $\ell^2(\mathbb{N})$ with coordinate basis $\{\phi_j\}$. The drift is $\mathcal{A} = \mathcal{A}_0 + \mathcal{K}$, where $\mathcal{A}_0\phi_j = -(1 + 0.01j^2)\phi_j$ and $\mathcal{K}\phi_1 = \sum_{j\geq 2} 12\, j^{-3}\phi_j$ (nilpotent, compact Volterra with $\rho(\mathcal{K}) = 0$). The noise is $\mathcal{D}\phi_j = j^{-3/2}\phi_j$, which is trace-class. The regularized conditioning acts at $k = 1$ with $\varepsilon_0 = 1$. Truncated covariances are compared against a finite reference $N_{\mathrm{ref}} = 1500$.

Table 1. Truncation convergence diagnostics for Experiment C.1.

Diagnostic	Value	Expected
$\|\mathcal{D} - \Pi_n\mathcal{D}\Pi_n\|_{\mathcal{S}_1}$ slope	-0.499	$\to 0$
$\|\Sigma_n - \Sigma_{\mathrm{ref}}\|_{\mathcal{S}_1}$ slope	-2.683	$\to 0$
$\|\Sigma_{n,\lambda} - \Sigma_{\mathrm{ref},\lambda}\|_{\mathcal{S}_1}$ slope	-2.683	$\to 0$
$\lvert S_{1,n,\lambda} - S_{1,\mathrm{ref},\lambda}\rvert$ slope	-5.504	$\to 0$
$m_2^{\mathrm{Schur},(1000)}$ (extrapolated)	0.5000000	0.5
$\lvert m_2^{\mathrm{Schur},(1000)} - \varepsilon_0/2\rvert$	6.87×10^{-14}	0

D.2. **Exact Variance Normalization Diagnostic.** We validate the exact variance normalization of Proposition 2 in its decoupled reference channel. With incoming arrows cut and off-diagonal drift coupling set to zero, the target covariance is exactly $\Sigma_{\lambda,22} = d_\lambda/(2\lambda)$. Four candidate target-diffusion families are tested:

(1) **Canonical:** $d_\lambda = \varepsilon_0/\lambda$.
(2) **Slow:** $d_\lambda = \varepsilon_0$.
(3) **Fast:** $d_\lambda = \varepsilon_0/\lambda^2$.
(4) **Perturbed:** $d_\lambda = (\varepsilon_0/\lambda)(1 + \lambda^{-2})$.

All evaluations are performed in Julia using 256-bit `BigFloat` arithmetic. A probe at $\lambda = 2^{100}$ resolves the nonzero finite-scale defect of the perturbed family.

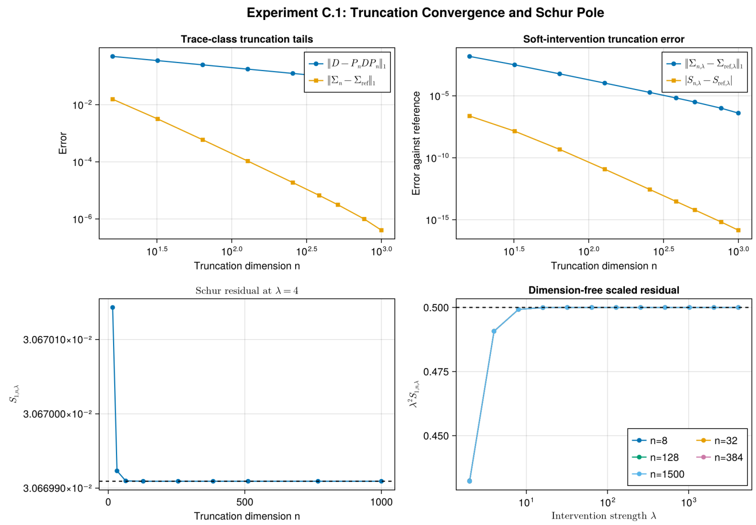

FIGURE 1. Experiment C.1. Trace-norm and residual errors decay in n; the extrapolated $m_2^{\mathrm{Schur},(1000)}$ differs from $\varepsilon_0/2$ by 6.87×10^{-14}.

TABLE 2. Exact-calibration ablation. Only the canonical family satisfies the exact finite-λ covariance condition.

Family	$\lambda d_\lambda/\varepsilon_0$	defect $\lvert 2\lambda^2\Sigma_{\lambda,kk}/\varepsilon_0 - 1\rvert$
Canonical	1	0
Slow	λ	$\lambda - 1 \to \infty$
Fast	λ^{-1}	$\lvert 1 - \lambda^{-1}\rvert \to 1$
Perturbed	$1 + \lambda^{-2}$	λ^{-2}

D.3. **Gaussian Finite-Part Functional in a Three-Node Chain.** We evaluate the Gaussian renormalized finite-part functional $\mathfrak{J}_\lambda$ on a three-node linear Ornstein–Uhlenbeck SDE:

$$X_1 \to X_2 \to X_3, \tag{107}$$

where X_2 is the target coordinate subject to regularized conditioning of strength $\lambda \in \{2^1, \ldots, 2^{12}\}$, with $x_2^* = 0$ and $\varepsilon_0 = 1$. The drift and diffusion operators are:

$$A_\lambda = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -\lambda & 0 \\ 0 & a_{32} & -1 \end{pmatrix}, \qquad D_\lambda = \mathrm{diag}(1, \varepsilon_0/\lambda, 1). \tag{108}$$

The target mean forcing b_2 controls the target mean shift $m_{\lambda,2} = b_2/\lambda$. We evaluate three Gaussian cases:

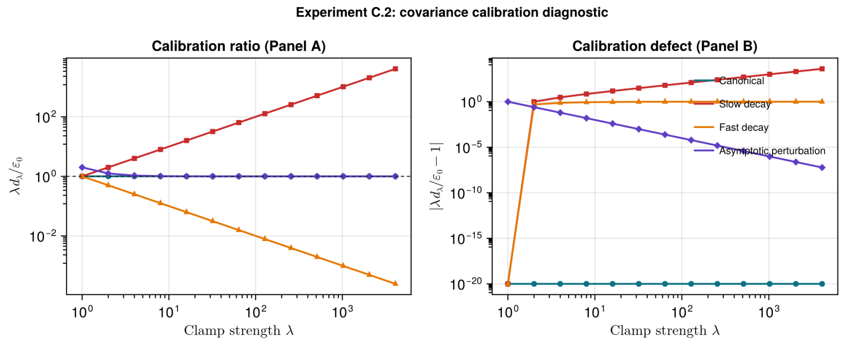

FIGURE 2. Experiment C.2. The perturbed family has the correct limiting ratio but fails the exact finite-λ covariance condition. Exact zero defects are displayed at 10^{-20}.

(1) **Case A (Decoupled-centered):** $a_{32} = 0, b_2 = 0$. Here $\mathfrak{J}_\lambda = 0$ exactly for all λ.
(2) **Case B (Coupled-centered):** $a_{32} = 1, b_2 = 0$. The coupling causes leakage $\Sigma_{23} \neq 0$, but the centering gives $\mathfrak{J}_\lambda = O(\lambda^{-4}) \to 0$.
(3) **Case C (Coupled-off-centered):** $a_{32} = 1, b_2 = 1$. The deterministic forcing causes mean shift $m_{\lambda,2} = 1/\lambda$, yielding $\mathfrak{J}_\lambda \to b_2^2/\varepsilon_0 = 1$.

All evaluations are performed in Julia using 256-bit `BigFloat` arithmetic, showing that the Schur residual $\lambda^2 S_{2,\lambda}$ converges to 1/2 in all cases, while $\mathfrak{J}_\lambda$ exhibits the exact zero, $O(\lambda^{-4})$ decay, and convergence to 1 behavior respectively.

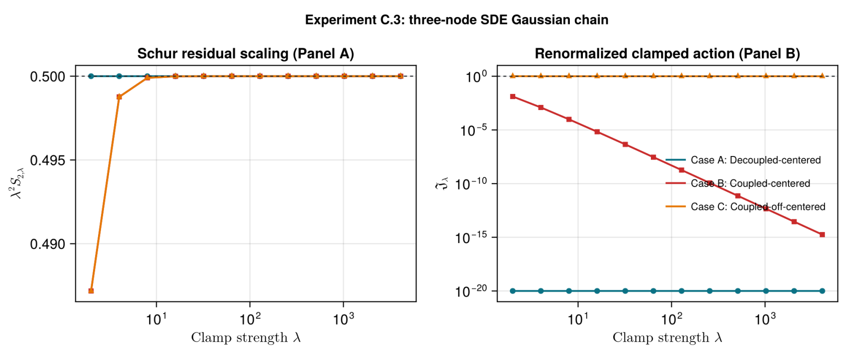

FIGURE 3. Experiment C.3. The panels show $\lambda^2 S_{2,\lambda} \to 0.5$ and the three finite-part regimes: exact zero, $O(\lambda^{-4})$ decay, and convergence to 1. Exact zeros are displayed at 10^{-20}.

D.4. **Numerical Precision Audit.** At $n = 256$, we recompute representative C.1 Schur residuals using IEEE 754 `Float64` and 256-bit Julia `BigFloat` arithmetic with the same covariance equation and Schur solve.

TABLE 3. Precision audit at $n = 256$; S_{256} uses 256-bit `BigFloat`.

λ	$\kappa_2(\Sigma_{II})$	$\lvert\lambda^2 S - \frac{1}{2}\rvert$	$\lvert S_{64} - S_{256}\rvert/\lvert S_{256}\rvert$
4	2.97×10^{6}	3.45×10^{-1}	6.50×10^{-16}
64	9.14×10^{5}	1.13×10^{-3}	3.48×10^{-17}
4096	9.14×10^{5}	9.20×10^{-10}	5.54×10^{-17}

For the five-point extrapolation of $m_2^{\mathrm{Schur},(256)}$, `Float64` and `BigFloat` differ by 9.98×10^{-17}; finite-λ remainder, not rounding, governs the observed distance to the limiting constant.

Remark 27 (Block-decoupling error suppression). The sub-machine-epsilon agreement between `Float64` and `BigFloat` in Table 3 is not coincidental; it is a direct consequence of the block-triangular structure established in this paper. Under the block-decoupled conditioning, the drift matrix takes the form

$$\mathcal{A}^{(\lambda)} = \begin{pmatrix} -\lambda & 0 \\ a_{Ik} & A_{II} \end{pmatrix}, \tag{109}$$

so the operator Lyapunov equation decomposes into three sub-problems:

(1) The target block $\Sigma_{\lambda,kk} = d_\lambda/(2\lambda)$ is solved by a single scalar division, incurring at most $\frac{1}{2}\epsilon_{\mathrm{machine}}$ relative rounding error (IEEE 754 round-to-nearest).
(2) The cross-covariance $\Sigma_{\lambda,Ik}$ satisfies $|\Sigma_{\lambda,Ik}|_{Q_0} = O(\lambda^{-3})$ by (16), so the absolute rounding error in the cross block is $O(\lambda^{-3}\epsilon_{\mathrm{machine}})$.
(3) The free block $\Sigma_{\lambda,II}$ is a standard Lyapunov solution with absolute error $O(\epsilon_{\mathrm{machine}}\|\Sigma_{\lambda,II}\|)$.

When the Schur complement is assembled as $S_{k,\lambda} = \Sigma_{\lambda,kk} - R_\lambda$, the subtracted variational term satisfies $R_\lambda = O(\lambda^{-6})$ (Theorem 9), which is negligible compared to $\Sigma_{\lambda,kk} = O(\lambda^{-2})$. The rounding error in $S_{k,\lambda}$ is therefore dominated by the error in $\Sigma_{\lambda,kk}$ alone, yielding a Schur-level rounding budget of $O(\epsilon_{\mathrm{machine}} \cdot \Sigma_{\lambda,kk})$ rather than the naïve $O(\epsilon_{\mathrm{machine}} \cdot \|\Sigma_{\lambda,II}^{-1}\| \cdot \|\Sigma_{\lambda,Ik}\|^2)$ that one would expect without block decoupling.

In short, the block-triangular structure converts what would be a catastrophically ill-conditioned Schur computation into a well-conditioned scalar computation plus negligible corrections. The observed `Float64`–`BigFloat` agreement at $O(10^{-17})$ is the numerical signature of this structural decoupling.

D.5. **The dichotomy of dual geometries.** To illustrate the simultaneous collapse and survival of the unified geometry under singular conditioning, we track the target scaling metrics on a 3-node serial OU chain under target coordinate conditioning on X_2 ($\varepsilon_0 = 1$). The 0th-order normal metric is extracted via the pull-back Jacobian:

$$g_{\lambda,\mathrm{normal}} := \lambda^{-2} S_{K,\lambda}^{-1}. \tag{110}$$

In a log-log evaluation over $\lambda \in [2^1, 2^{20}]$, the normal metric $g_{\lambda,\mathrm{normal}}$ instantly stabilizes to the horizontal plateau $2/\varepsilon_0$. Conversely, the unrenormalized horizontal lift response $g_\lambda^{\mathrm{bulk}} = A_\lambda^{-2} R_\lambda$ collapses strictly at a slope of -2 ($O(\lambda^{-2}) \to 0$). Upon RFIM rescaling (λ^2), the 1st-order metric locks onto a distinct constant offset perfectly matching the theoretical Cameron–Martin limit $\|a_{Ik}\|^2_{Q_0^{-1}}$.

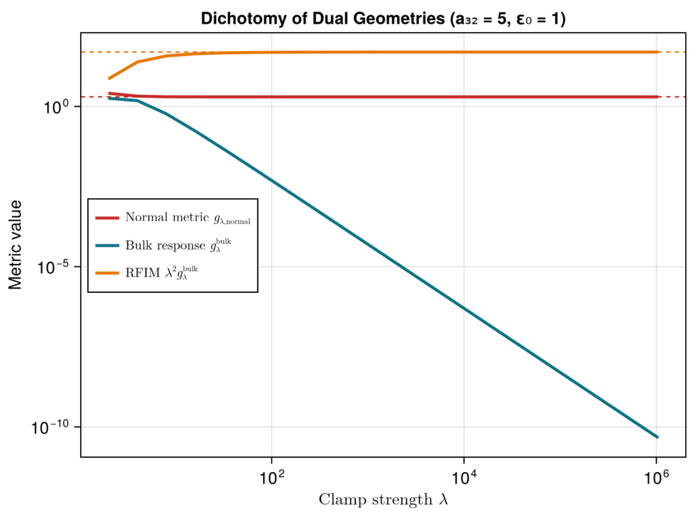

FIGURE 4. Dichotomy of dual geometries for the 3-node serial Gaussian chain under target coordinate conditioning on X_2 ($\varepsilon_0 = 1$, $a_{32} = 5$). The 0th-order normal metric $g_{\lambda,\mathrm{normal}}$ (red) instantly stabilizes at the horizontal plateau $2/\varepsilon_0 = 2$, showing the survival of the boundary geometry. The unrenormalized horizontal lift response $g_\lambda^{\mathrm{bulk}}$ (blue) collapses strictly at a slope of -2 as $O(\lambda^{-2})$. Rescaling the bulk response by λ^2 yields the 1st-order metric (orange), which locks onto the non-zero Cameron–Martin limit $\|a_{Ik}\|^2_{Q_0^{-1}} = 2a_{32}^2 = 50$.

The log-log scaling plot in Figure 4 demonstrates that while the horizontal connection collapses, the boundary normal geometry survives intact, and the renormalized 1st-order perturbation retains the coupling invariant.

D.6. **Orthogonal independence of normal geometry and horizontal lift.** To verify that the 0th-order boundary normal metric and the 1st-order horizontal lift quantify logically independent responses, we preserve the exact target scaling ($\varepsilon_0 = 1$) while toggling the outgoing cross-coupling a_{32}. As demonstrated in Table 4, modifying the off-diagonal leakage strictly varies the 1st-order residual while leaving the normal geometry universally preserved.

TABLE 4. Verification of orthogonal decoupling of normal geometry and horizontal lift

Environment	Coupling a_{32}	0th-order Metric	1st-order Metric (RFIM)
Decoupled	0	2.00000000	0.00000000
Coupled	5	2.00000000	49.99990463

Table 4 numerically proves the conceptual core: the 0th-order normal metric is uniquely calibrated by the unperturbed evidence noise scale, while the 1st-order horizontal lift is exclusively governed by the off-diagonal Cameron–Martin connection energy.

D.7. **Convergence to the Mahalanobis normal well.** Because the background regularized-conditioning measure constitutes a Gaussian location family with covariance invariant to the evidence offset b, the blow-up Fisher metric matrix $G_{\lambda,K} := \lambda^{-2} S_{K,\lambda}^{-1}$ remains rigidly constant with respect to b at every finite λ, yielding a purely flat geometry ($\Gamma = R \equiv 0$). We verify the constant convergence rate of the matrix geometry in a target block of $|K| = 2$. As $\lambda \to \infty$, the finite-λ flat quadratic cost contours converge unconditionally to the limiting Mahalanobis well governed by the Schur second-order coefficient matrix E_0:

$$\mathfrak{D}_{\lambda}^{\mathrm{FP}}(b,0) = \frac{1}{2} b^{\top} G_{\lambda,K} b \longrightarrow \mathfrak{D}^{\mathrm{FP}}(b,0) = b^{\top} E_0^{-1} b. \tag{111}$$

Evaluation confirms that the distance between the intermediate geometries decays in Frobenius norm as $\|G_{\lambda,K} - 2E_0^{-1}\|_F = O(\lambda^{-4})$.

The contour comparison in Figure 5 (Panel A) demonstrates that the quadratic geometry of the blow-up divergence rapidly converges to the limiting Mahalanobis well. The error decay in Figure 5 (Panel B) confirms that the distance between the intermediate and limiting geometries in Frobenius norm decays strictly at the Schur geometric rate of $O(\lambda^{-4})$. Thus, the singularity does not generate curvature; it selects a finite flat metric. This flatness is not a vacuous triviality, but a structural consequence of the location family under regularized conditioning, with the hard-limit topological clamp rigorously preserving the displacement flatness.

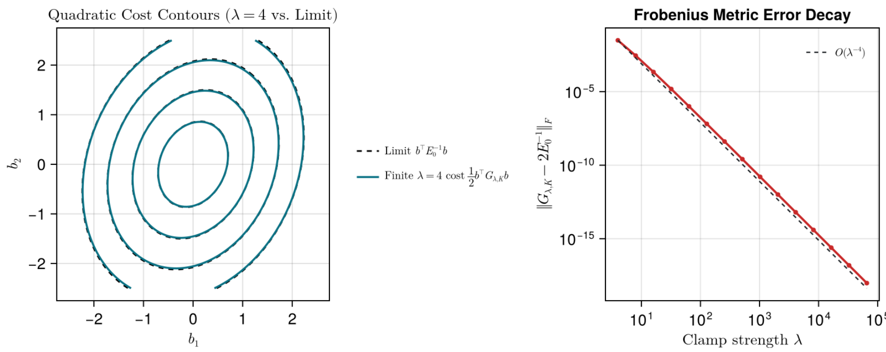

FIGURE 5. Convergence of the finite-λ quadratic cost contours to the limiting Mahalanobis normal well for target block $|K| = 2$ with off-diagonal coupling. **Left (Panel A):** Contours of the finite-λ cost $\frac{1}{2} b^\top G_{\lambda,K} b$ at $\lambda = 4$ (solid blue lines) overlaid with the contours of the limiting Mahalanobis cost $b^\top E_0^{-1} b$ (dashed black lines), demonstrating close alignment even at a low regularizer strength. **Right (Panel B):** Log-log plot of the Frobenius norm error $\|G_{\lambda,K} - 2E_0^{-1}\|_F$ (solid red line with markers) versus the regularizer strength $\lambda \in [2^2, 2^{16}]$, running in parallel with the theoretical $O(\lambda^{-4})$ slope reference (dashed black line).